Customer logo

Henkel

Functional Design Specification

SampleManager PhaseB

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Date	
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Document Signoff				
Name	Position	Company	Signature	Date

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1.2	Add chapter 3.2.1.5	Zhenjuan.Zhang	2025/07/04

2 Introduction

2.1 This Document

2.1.1 Document overview

This document describes the system functionality of Phase B pilot stage covered in the Henkel QCLad System RTM User Requirements Analysis Report.

2.1.2 Disclaimer

The screenshots used below are screenshots generated during the system standard functions or system configuration, as well as customer sample screenshots, the final design may be different from the screenshots in the document, and the data on the screenshots is not the real data of the system, only for reference.

2.1.3 Purpose

This document is used to provide a detailed description of the functionality of the SampleManager system, identifying a description of the system functionality that meets the user's business needs as specified in the phase B pilot stage of User Requirements Analysis Report. The functionality of phase A, which is already deployed in the system, will not be described further

2.1.4 Scope

- Document the functional specifications of the laboratory management system based on the requirements described in the User Requirements Analysis Report.

- Determine the implementation of functionality in the system based on the needs described in the user needs analysis report

2.2 Synopsis

The core application functionality of the QCLad system is the laboratory sample process management functionality, including sample management, SAP interface, task scheduling, master data, reporting, architecture and security.;

2.3 Definitions

The following abbreviations and definitions are assumed throughout this document.

Term/ Abbreviation	Meaning	Description
FS	Function Specification	This Document

Term/ Abbreviation	Meaning	Description
RMB	Right Mouse Button	Using RMB various Sample Manager options can be displayed
QCLAD	SampleManager	Software Name

2.4 References

The following references are made in this document.

Ref. No.	Reference	Description

3 Project Brief

3.1 Project Overview

3.1.1 Project detail

Implementation of SampleManager QCLad will streamline Sample Management, Reports, Dashboards, Instrument and Equipment management, Instrument interfacing and External application integration together with the associated data acquisition across Henkel's labs.

3.2 Functional Requirements

3.2.1 Sample Management

3.2.1.1 Lab Testing Process Workflow

3.2.1.1.1 Business Scenarios

Generally, samples and tests will flow the process of workflow which is deployed during project stage phase A. In this document it will be referred to as the *lab testing process* and will be used in the lab as a base process for other subsequent workflows.

3.2.1.1.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.5.1.1	System shall be able to manage Standard RM test for Incoming workflow. （Standard RM）	Configuration
4.1.5.1.2	System shall be able to manage Standard in process testing for semi finished and finished goods workflow. （without MES)	Configuration
4.1.5.1.3	System shall be able to manage Standard finished goods outgoing test workflow.	Configuration
4.1.5.1.10	System shall be able to manage Test for SCMS/NCMR /CCMS investigation workflow.	Configuration

3.2.1.1.3 Function Design

The *lab testing process* workflow defines the lifecycle of a sample from received status to authorization. See the flow chart below.

Samples and tests will be run according to this workflow if not otherwise specified.

3.2.1.2 Truck Workflow

3.2.1.2.1 Business Scenarios

There are TRUCK samples that need to be created manually first, and then the sap inspection batch will be pushed. QCLAD needs to be able to return the results correctly.

3.2.1.2.2 RTM Mapping

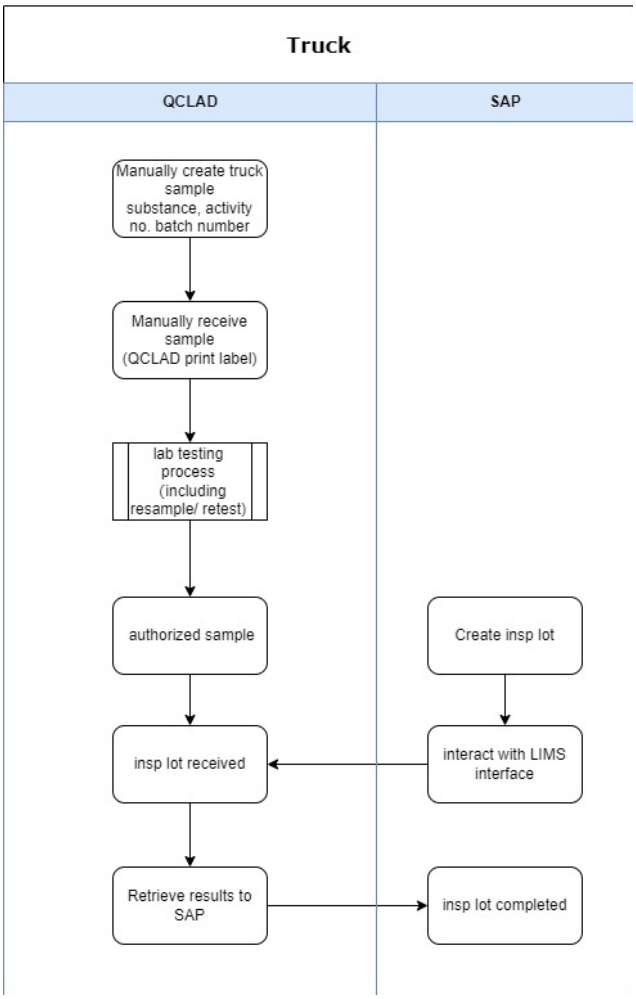
URS Ref.	Requirement	Type
4.1.1.1.1	System shall be able to login samples using configurable templates	Configuration
4.1.1.1.3	System shall be able to create samples manually for standard testing.	Configuration

3.2.1.2.3 Function Design

3.2.1.2.3.1 Design Overviews

Users can create Tank samples and tests manually. QCLAD shall retrieve the results to SAP after inspection lot is received. (Based on RTM as above).

In the *Sample management* cabinet, samples and tests will be displayed in subfolders categorized by sample type. Truck sample will be grouped in sub folder IQC-TRUCK.



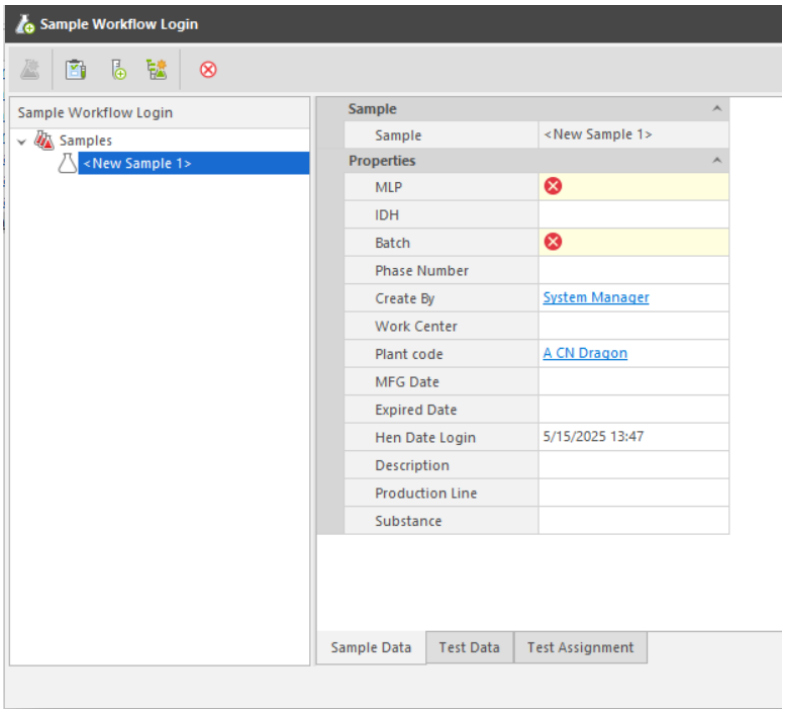
3.2.1.2.3.2 Process Design

Sample Login:

Truck Samples will be login by manually running sample login workflow *Truck Sample* by navigating to: Public folders-Sample Management-Create Sample.

The user needs to select the MLP and enter the batch number. QCLAD system will automatically fetch the IDH and activity number from the master data.

For truck material, creates the sample id by adding the entries.



Properties in entity template clarification in LIMS:

No.	Properties	Type	Prompt Type	Description
1	MLP	Identity	Mandatory	Links to MLP table
2	Batch	Text	Mandatory	Batch id
3	Plant code	Text	Optional	Links to Plant table
4	Activity No	Numeric	Optional	Fetch from MLP
5	Work Center	Text	Optional	Links to Work centre table
6	Date Login	Date	Optional	Current date, modifiable.
7	Create By	Operator	Optional	Current operator
8	Description	Text	Optional	Descriptiom
9	Substance	Identity	Optional	Fetch from MLP
10	MFG Date	Date	Optional	Manufacture date
11	Expired Date	Date	Optional	Expiration date
14	Production Line	Identity	Optional	Fetch from MLP

Sample Receiving:

The user needs to find the specified sample from the *New Sample Pending Receiving* folder and right-click *Receive Samples* button to open the Sample Receiving window. After the sample is received, the sample label is automatically printed to the label printer in the corresponding work centre.

Sample Authorize:

When all tests for a sample are in the authorized or rejected status, the sample will automatically become authorized.

Authorized JIT samples will be listed in the *Sample management-Authorized samples* folder. System will highlight the JIT samples which has not received inspection lots from SAP by highlighting the background color in the Authorized samples folder.

Truck Samples are not automatically marked as waiting to be sent results to SAP when the sample is authorized, but need to wait for a response from the sap interface.

3.2.1.2.3.3 SAP interactive

Truck sample is created manually in QCLAD system.

When inspection lot created in SAP is pushed to QCLAD, interface would find the sample already existed in system and save the inspection lot number in the field 'hen_inspection_lot' and mark the field 'hen_to_be_sent' to true.

3.2.1.3 JIT Workflow

3.2.1.3.1 ***Business Scenarios***

There are JIT samples that need to be created manually first. When all tests are all completed and reviewed, QCLAD will send the message which including sample decision information(A/R) to the MES. Then SAP will push the inspection lot and QCLAD needs to be able to return the results correctly.

3.2.1.3.2 ***RTM Mapping***

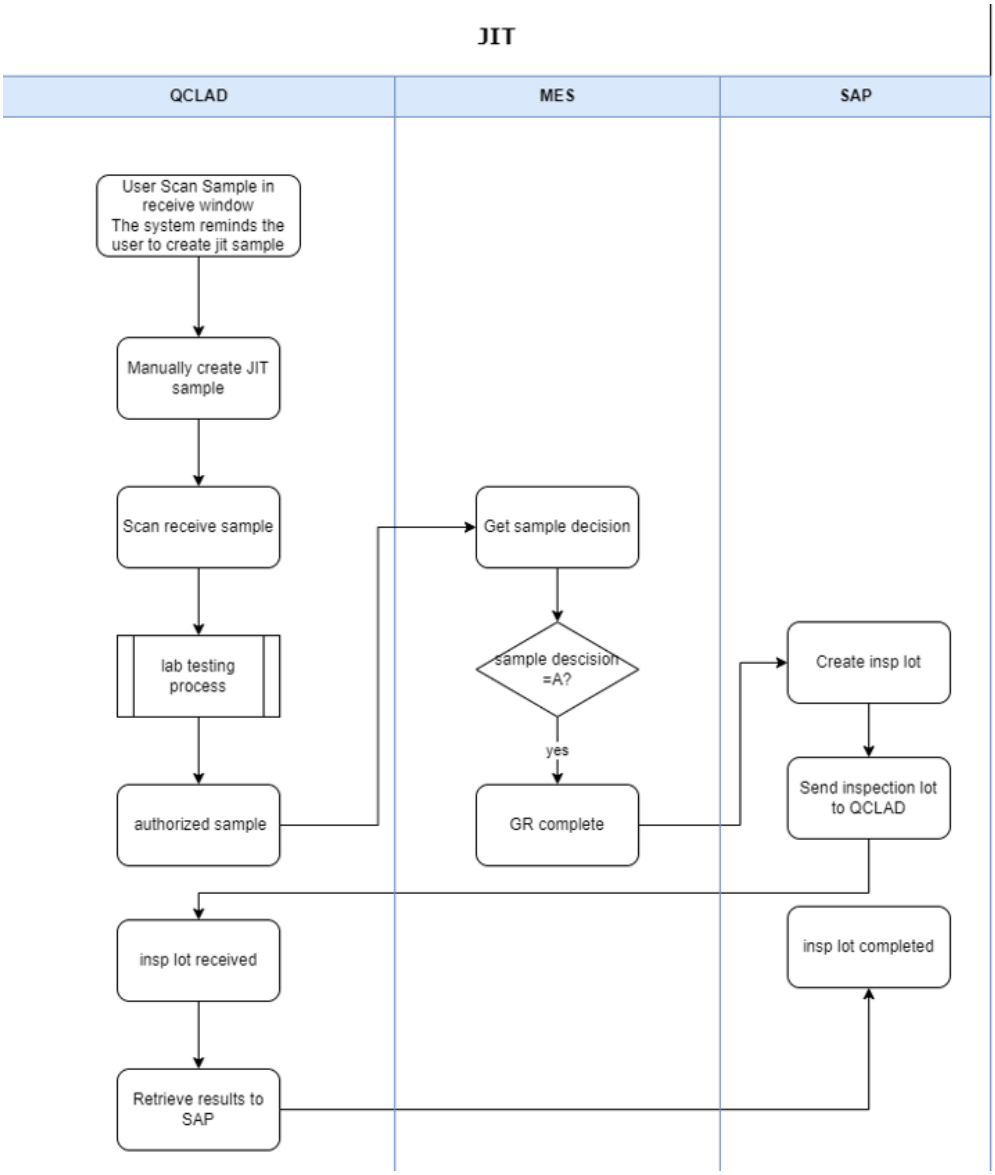
URS Ref.	Requirement	Type
4.1.1.1.1	System shall be able to login samples using configurable templates	Configuration
4.1.1.1.3	System shall be able to create samples manually for standard testing.	Configuration

3.2.1.3.3 ***Function Design***

3.2.1.3.3.1 Design Overviews

Users can create JIT samples and tests manually. QCLAD shall send the message to MES when sample is authorized and retrieve the results to SAP after inspection lot is received. (Based on RTM as above).

In the *Sample management* cabinet, samples and tests will be displayed in subfolders categorized by sample type. Truck sample will be grouped in sub folder IQC-JIT.



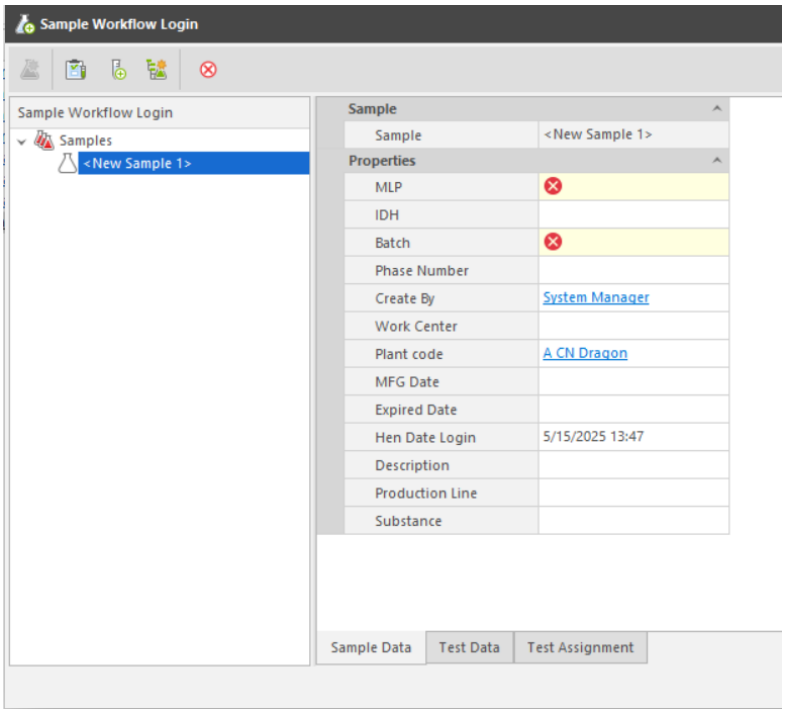
3.2.1.3.3.2 Process Design

Sample Login:

JIT Samples will be login by manually running sample login workflow *JIT Sample* by navigating to: Public folders-Sample Management-Create Sample

System shall remind user to create JIT sample when scanning receive the sample which has not yet been created in QCLAD

The user needs to select the MLP and enter the batch number. QCLAD system will automatically fetch the IDH and activity number from the master data.



Properties in entity template clarification in LIMS:

No.	Properties	Type	Prompt Type	Description
1	MLP	Identity	Mandatory	Links to MLP table
2	Batch	Text	Mandatory	Batch id
3	Plant code	Text	Optional	Links to Plant table
4	Activity No	Numeric	Mandatory	Activity number
5	Work Center	Text	Optional	Links to Work centre table
6	Date Login	Date	Optional	Current date, modifiable.
7	Create By	Operator	Optional	Current operator
8	Description	Text	Optional	Descriptiom
9	Substance	Identity	Optional	Fetch from MLP

10	MFG Date	Date	Optional	Manufacture date
11	Expired Date	Date	Optional	Expiration date
14	Production Line	Identity	Optional	Fetch from MLP

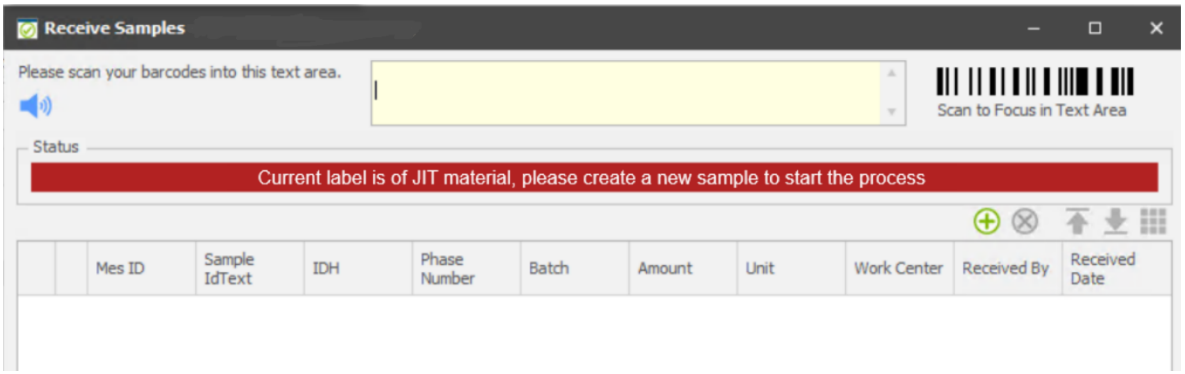
Sample Receiving:

Users can receive sample by scanning the MES label in sample receive window or manually finding the specified sample from the *New Sample Pending Receiving* folder.

After the sample is received successfully, the sample label is automatically printed to the label printer in the corresponding work centre.

The MES label should have a QR code that contains the information in the same format as phase A.

When a user scans to receive a JIT sample but the sample has not yet been created in QCLAD, system will remind the user in the scan window that the sample is a JIT sample.



Sample Authorize:

When all tests for a sample are in the authorized or rejected status, the sample will automatically become authorized.

Authorized JIT samples will be listed in the *Sample management-Authorized samples* folder. System will highlight the JIT samples which has not received inspection lots from SAP by highlighting the background color in the Authorized samples folder.

JIT Samples are not automatically marked as waiting to be sent results to SAP when the sample is authorized but need to wait for a response from the sap interface.

3.2.1.3.3.3 SAP interactive

JIT sample is created manually in QCLAD system.

When inspection lot created in SAP is pushed to QCLAD, interface would find the sample already existed in system and mark it as waiting to be sent results.

3.2.1.3.3.4 MES interactive

When the sample is authorized, QCLAD determines the sample decision based on the *on spec* field on the sample table and sends a string to the MES interface.

3.2.1.4 Retain Samples Workflow

3.2.1.4.1 ***Business Scenarios***

The laboratory will retain authorized samples for a certain period of time in the sample retention room. Samples may be taken out for testing during the retention period. The samples will be scrapped after the expiry date.

3.2.1.4.2 ***RTM Mapping***

URS Ref.	Requirement	Type
4.1.6.1.2	System shall be able to support QC retain sample, consumables and spare parts can select location.	Configuration
4.6.1.16.1	System shall display retain sample in dashboard by storage location, retain status , number of expired retains, disposed retains by date.	Customization
4.8.5.3.1	System shall allow user to record and retrieve the storage location of samples stored in QC by providing material, batch information.	Configuration

3.2.1.4.3 ***Function Design***

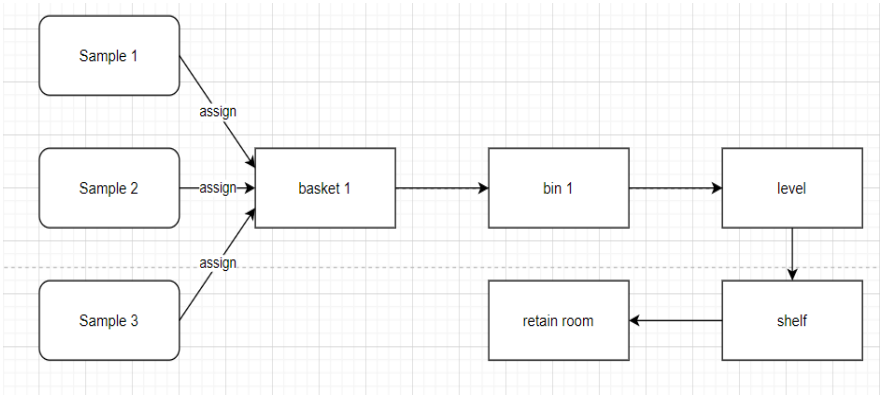
3.2.1.4.3.1 Design Overviews

Users can trigger sample retain by click on retain sample button or right click on samples, scanning sample labels in the barcode scanning window.

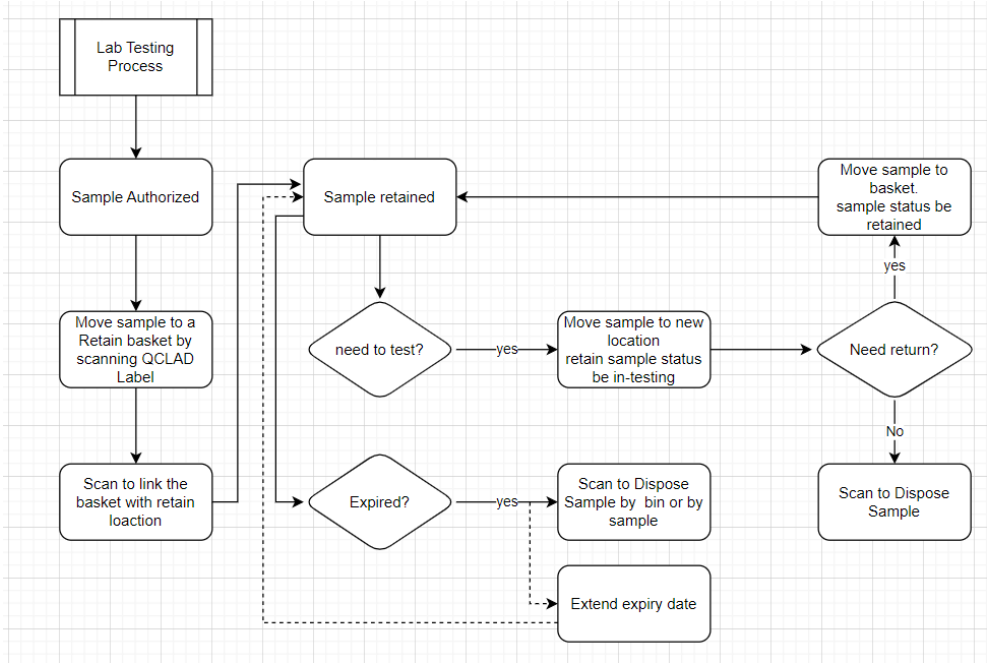
When retaining a sample, User can assign it to a lab basket. And the baskets can be assigned to bins.

Users can scrap the retain samples by scanning the bin or basket label to scrap by bin or scanning sample label to scrap by sample.

Retain location hierarchy:



Retain sample workflow:



3.2.1.4.3.2 Process Design

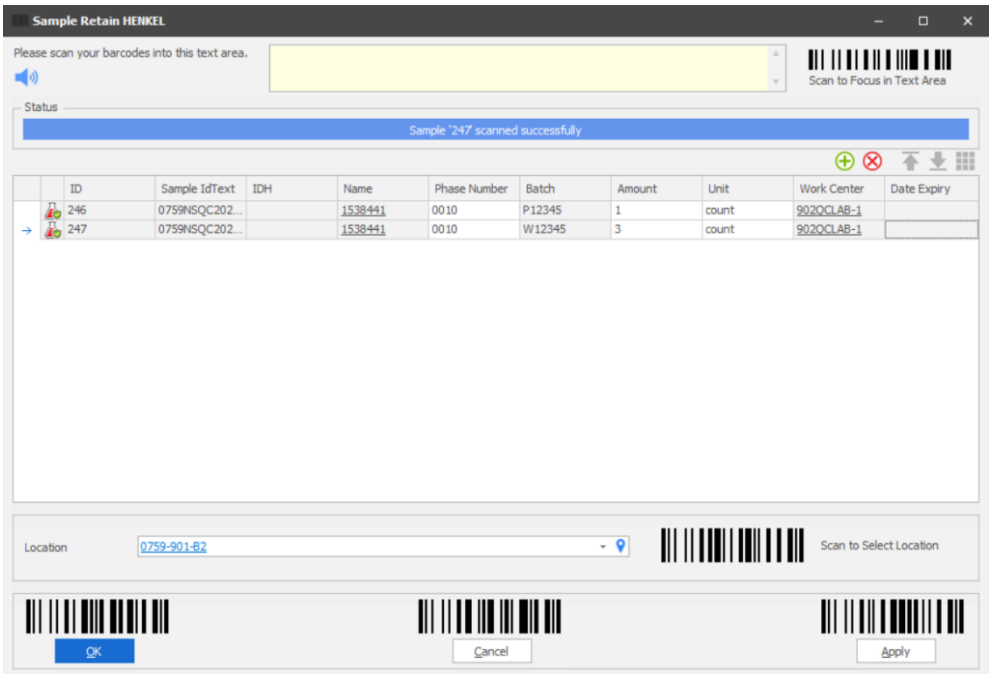
Sample Retaining:

User can trigger retain scanning process by clicking *retain sample* button.

User shall scan the QCLAD label on the sample and select the location by scanning the label on basket to retain sample in the basket. If there is no basket concept at plant level, user can scan the bin QR code.

If the bin is set up with a work center limit, then only samples belonging to that work center are allowed to be placed in it.

System shall record the retain date, retain amount(optional) and operator.



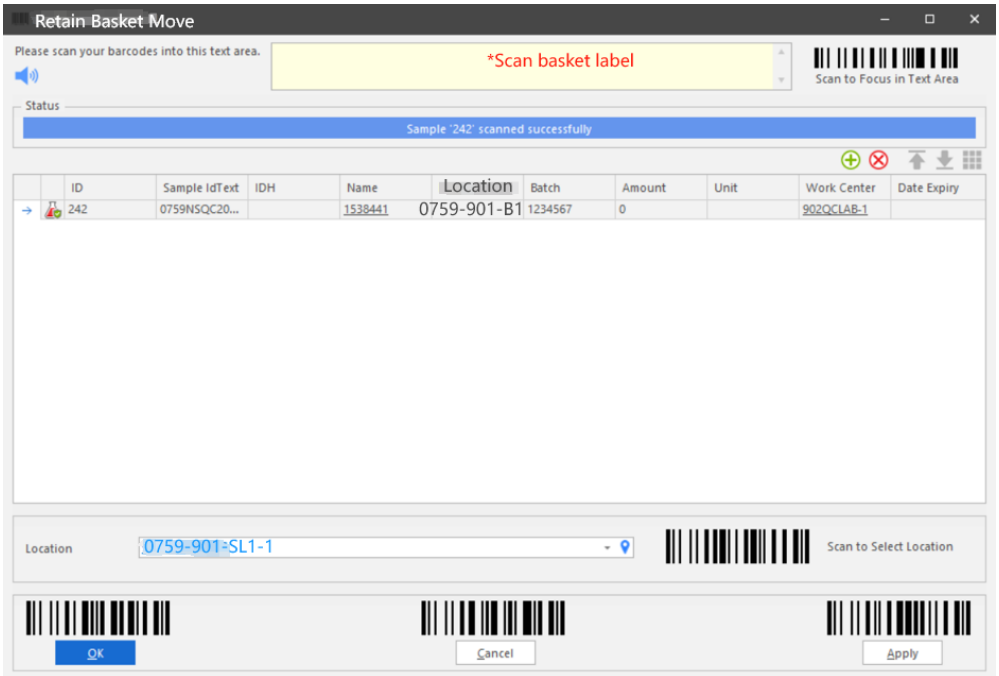
Click OK or Apply to retain the sample in the basket.

Basket Move Location

Each Basket kept in the bin has its own parent location, the bin.

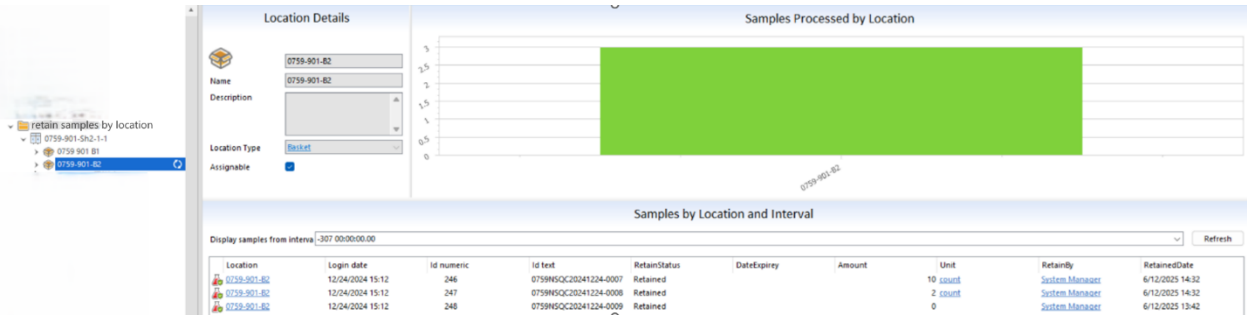
Baskets listed can be found in retain baskets folder where bin information is displayed.

If user wants to assign the basket to a new bin, user can click the *Move Basket to new Bin* button and scan the basket label. Then scan the bin label in location field to move.



Retain Query

Retain samples will be listed by location and basket in *retain samples by location* folder.



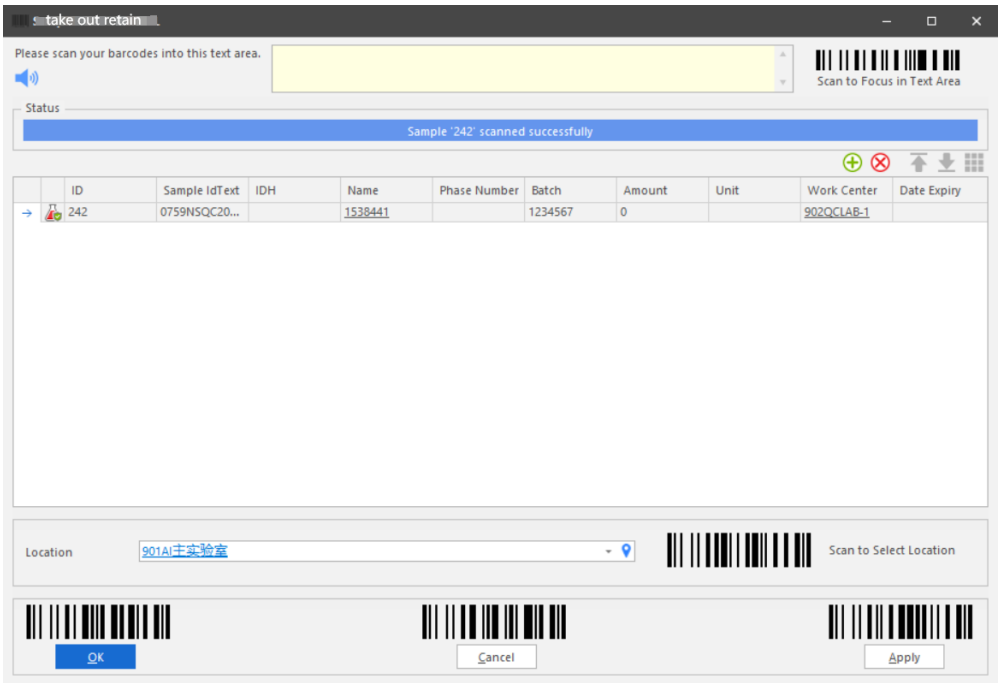
The Expired retain samples will be displayed in the *expired retains* folder

The Bins will display the oldest expire date from all its stored samples in the *bin list* folder.

The Bins whose retain samples stored are all expired will be displayed in *to-be scrapped* folder.

Retain Sample Take-out

User can take out the whole retain sample for other use by clicking button *take out retain*. User shall assign a new location to take out the sample. Once the retain sample is taken out, its retain status would be *take out*.

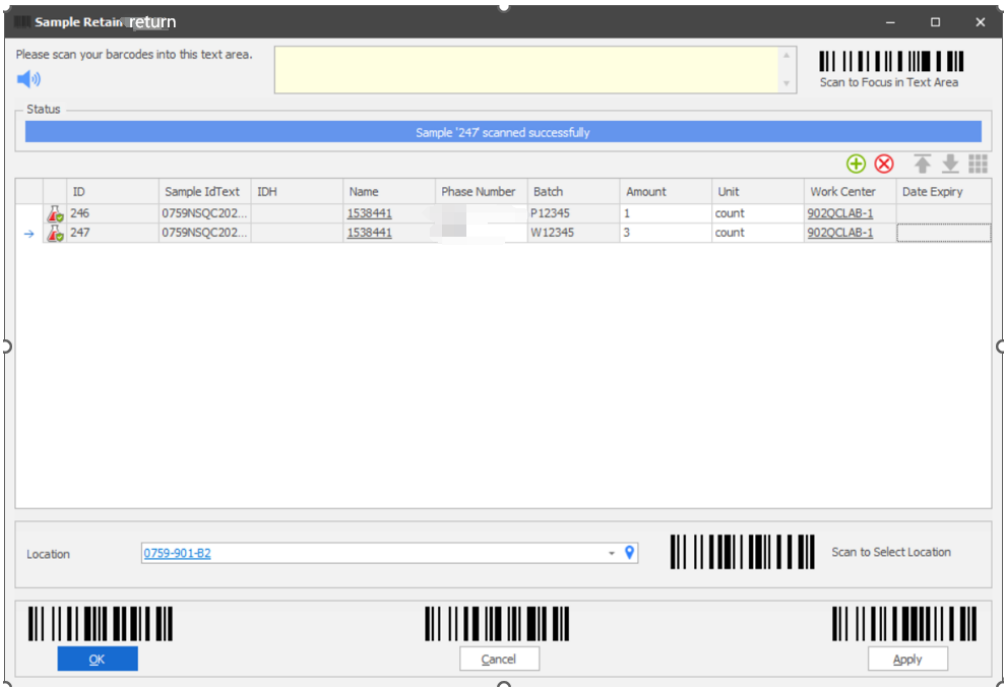


After usage of retain sample, user can return the sample to retained by *return retain sample* button or scrap the sample by *scrap retain* button.

Retain Sample Return

After a retain sample is taken out, if user wants to return it to retain storage, User shall click *return retain sample* button. Scan the sample label and basket label and record return amount.

User can modify the expiry date in the sample return scanning window. The expiry date is a mandatory field when returning retain samples.Once the retain sample is returned, its retain status would be *retained*.

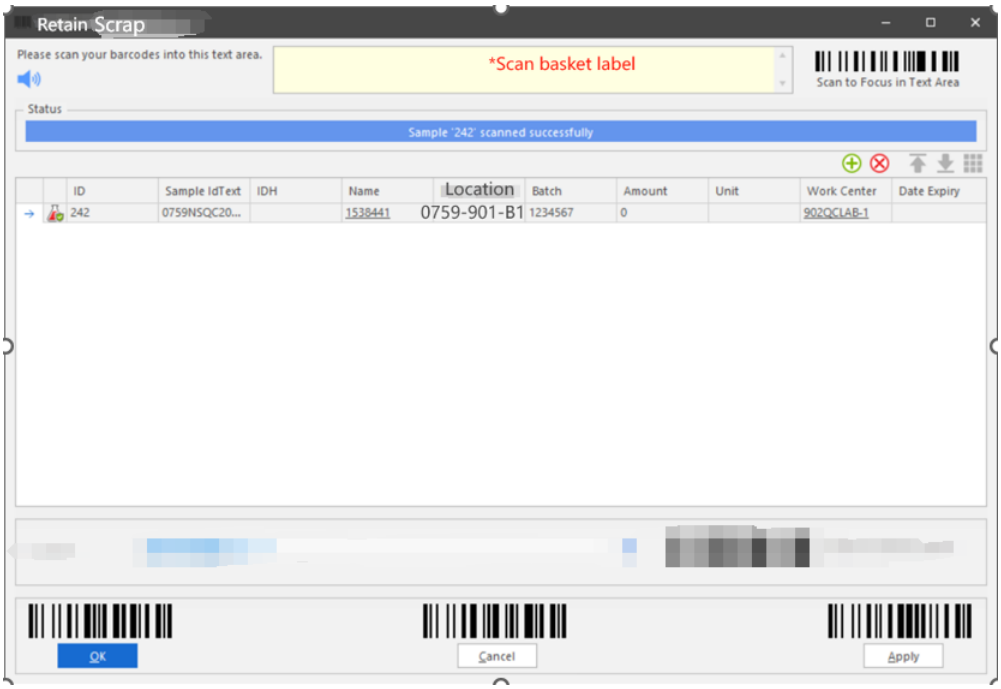


Retain Scrap

User can scrap the retain sample by samples by sample or by bin.

If the bin method is used, all samples in the bin will be scraped together by scanning basket label.

System shall record the scrap amount, scarp date and operator. And the retain sample status would be ‘scrapped’. The scrapped samples will be displayed in scraped retain folder.



Retain Expiry Date Extension

User can extend the retain expiry date by 2 ways:

1. When a retain sample is returned, the expiration date can be modified in the return scanning window. The expiration date field is mandatory when returning the retain sample.
2. User can use *modify expiry date* button on retain samples or scan the sample label to modify expiry date.

3.2.1.5 Consumables Workflow

3.2.1.5.1 ***Business Scenarios***

When stock is put into storage, some stock types need to be tested. Only after the test sample is authorized can consumable be released for use.

User will consume certain stock batch when performing testing.

3.2.1.5.2 ***RTM Mapping***

URS Ref.	Requirement	Type
4.1.5.1.4	System shall be able to manage Consumables incoming test workflow.	Customization
4.1.6.1.2	System shall be able to support QC retain sample, consumables and spare parts can select location.	Configuration

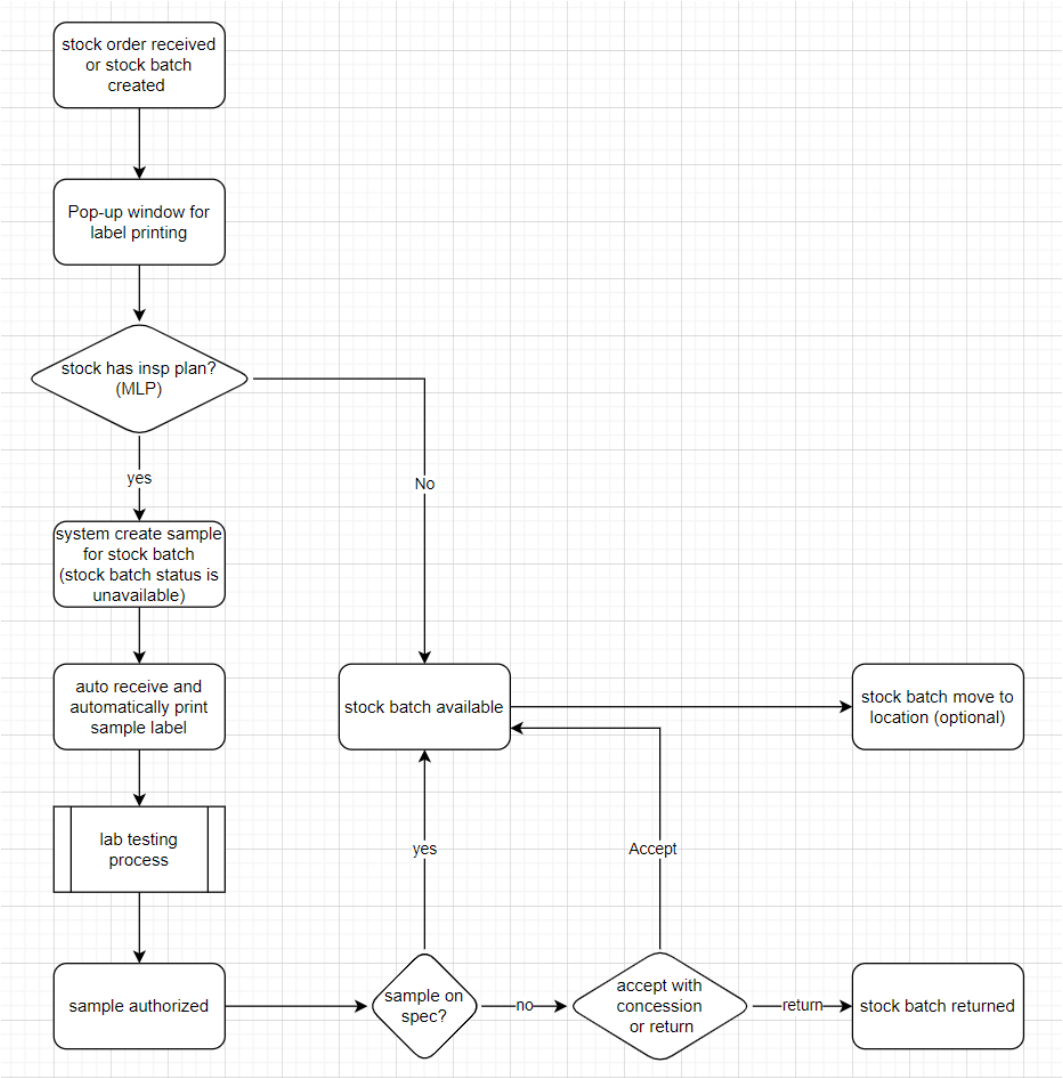
3.2.1.5.3 ***Function Design***

3.2.1.5.3.1 Design Overviews

Consumables are maintained in the system in the Stock module. When a stock is set up to require an inspection plan (MLP), QCLAD will automatically generate a test sample when that stock batch is created.

Until this sample is authorized, the stock batch will be in an unavailable status.

Once the sample is approved, the system will automatically release the inventory lot based on whether the sample passes or fails its specification, or user can manually release it.



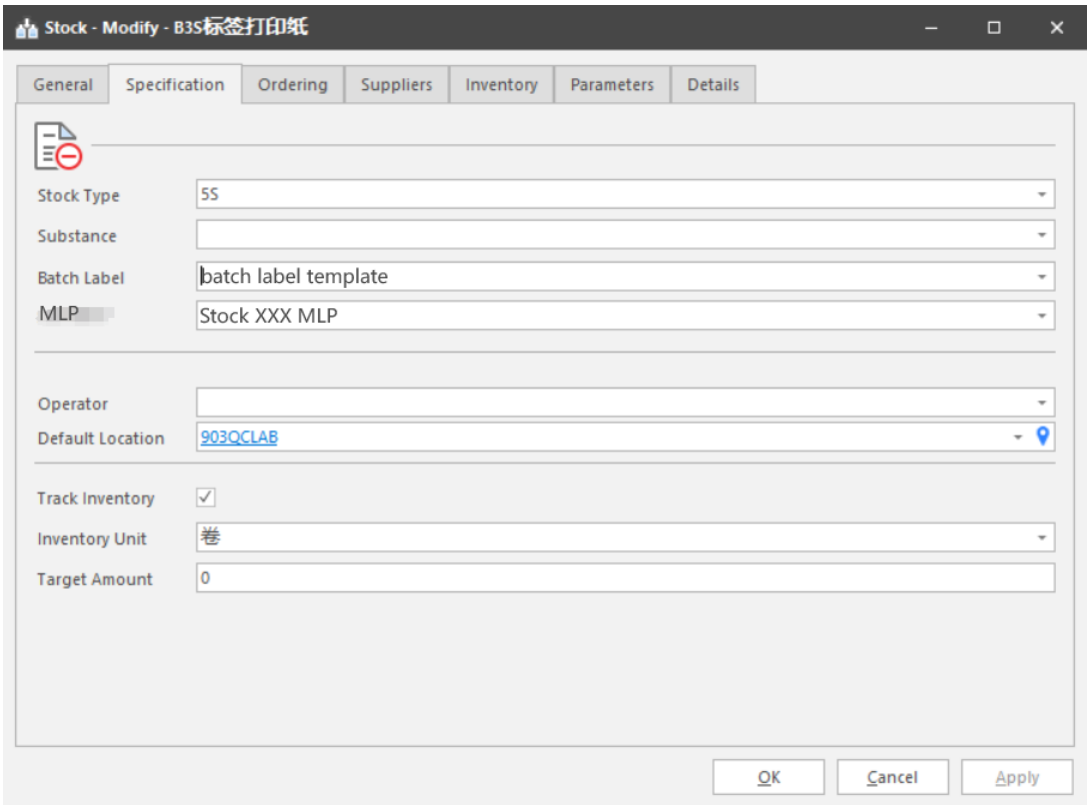
3.2.1.5.3.2 Process Design

Master Data

MLP information should be maintained in the STOCK master data, similar to the master data for other routine samples, where test schedules and corresponding specification information are associated with this MLP. The system will use this MLP to create test samples.

The label template that defines the label design that can be created for stock batches associated with the stock.

The Auto Consumption checkbox is used to determine if this stock can be used to automatic stock consumption during testing.



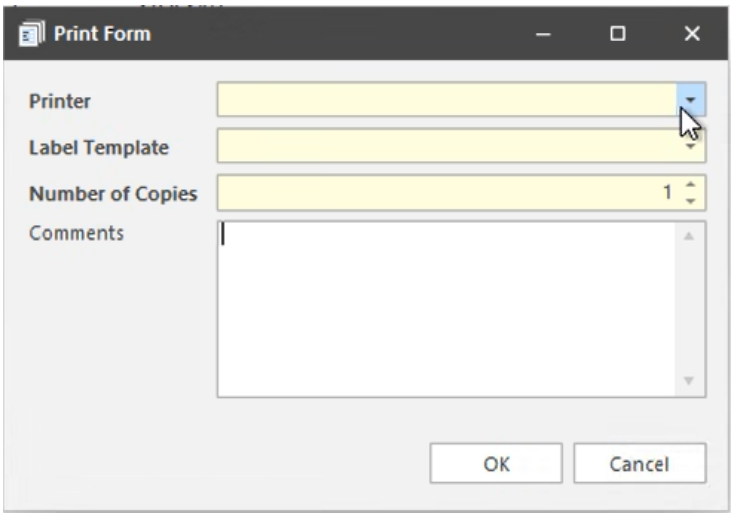
Stock Batch Creation:

When creating stock batches, Expiry date and batch number is mandatory if its stock is checked '*Auto Consumption*' in master data.

If a stock batch is created by stock order, expiry date and batch number are mandatory when receiving order. Group information shall be fetched from stock master data.The group shall be mandatory and allow user to modify it.

When a stock batch is created in QCLAD system, by stock order receiving or directly adding stock batch, the system will pop up a window asking for printing label quantity.

By default, the label quantity is 1 and the printer is the work center printer. Label template is the label template defined in stock master data. User can modify the number of printing labels or label template or another QCLAD printer.



Sample Login:

When a stock batch is created in QCLAD system, by stock order receiving or directly adding stock batch, the system will check its stock master data to determine if there is MLP selected for that stock.

If there is, then a sample will be created automatically according to the MLP, its sample type would be IQC-CON and its work center will be pulled from substance master data. The stock batch will be displayed in *stock batches unavailable* folder in *Stock Management* cabinet;

If not, then the stock batch will be directly released to available status.

Sample Process:

The consumable samples will be listed in samples pending receiving folder. When user receviing the sampe, A sample label will be automatically printed by the work center printer.

Users can complete testing of the sample according to the Lab Testing Process Workflow in *Sample Management* cabinet folders.

Stock Batch Release:

When all tests of the sample are authorized, the sample get authorized:

If all tests under that sample meet the specification criteria, the system will automatically release the stock batch. Its status will become available;

If any of the tests under that sample do not meet the criteria, the system will not automatically release the stock batch. Its status will become *Await Decision.* User can manually click right mouse button to decide whether to return or release.

All returned stock batches are displayed in the *Returned Stock Batches* folder.

User can move the stock batch to its storage location using *Move Batch* button.

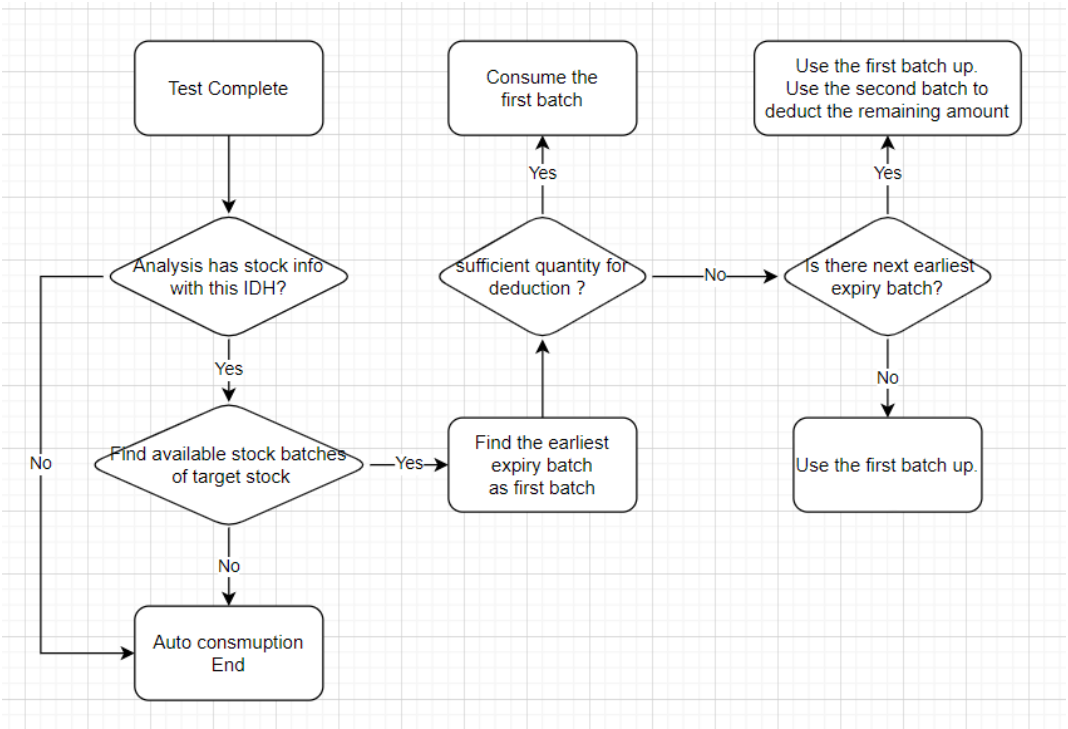
Automatic Consume Stock Batches to a Test:

To enable automatic stock consumption during testing, the following conditions must be met:

1. In Analysis master data, stock, consumption quantities and unit are required for the corresponding IDH.

2. There are available stock batches corresponding to analysis.

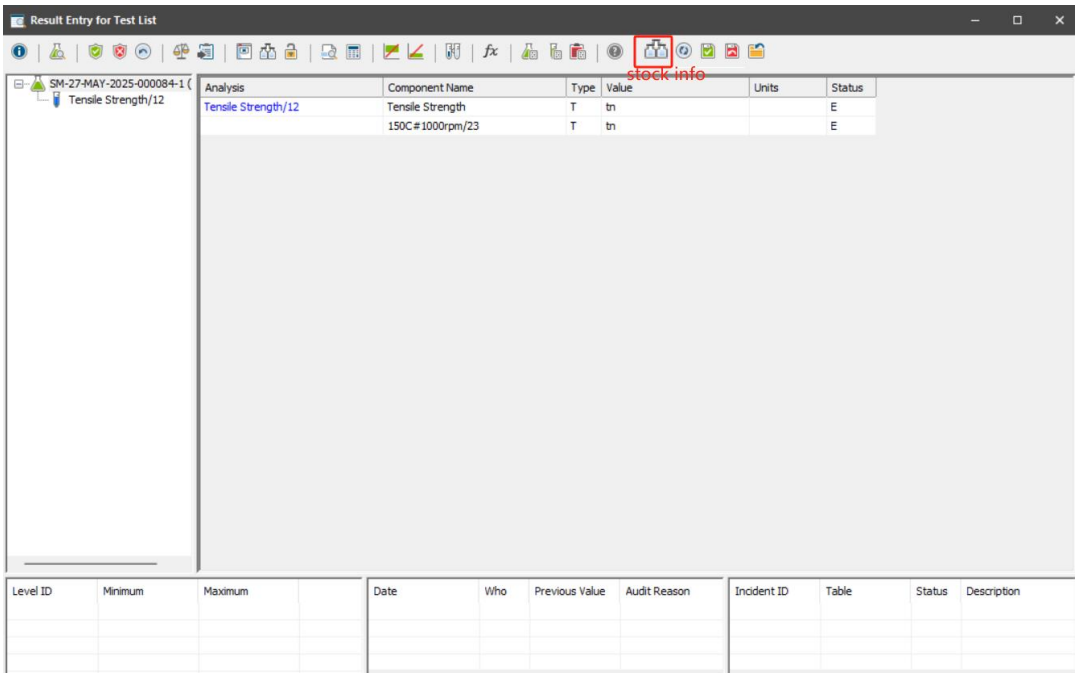
The automatic consumption deduction logic is shown in the flowchart below.



If the actual deduction amount is less than the target amount during the automatic deduction process, the system will pop up a window to remind the user of the relevant information.

Modifying Stock Batch Consumed to a Test

Users can click the newly added button on the Result Entry Window to open the Stock consumption window.



In the stock consumption window, system will list the stock batch and the amount consumed.

User can modify the stock batch or the consumed amount in the stock consumption window only when test has not been authorized.

3.2.1.6 Instrument Calibration

3.2.1.6.1 ***Business Scenarios***

Some laboratory instruments require regular calibration, which is divided into internal calibration and external calibration. Internal calibration require users to conduct tests according to preset testing items and determine whether the instrument is passed..

3.2.1.6.2 ***RTM Mapping***

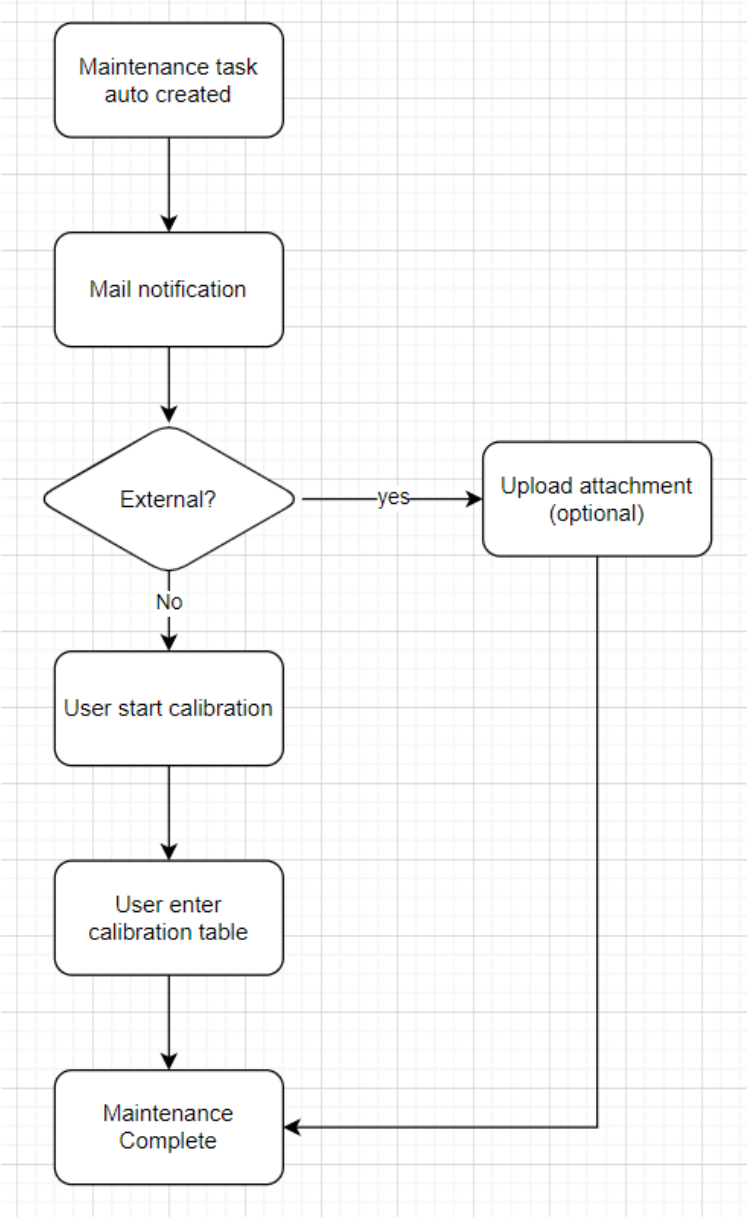
URS Ref.	Requirement	Type
4.8.1.1.1	System shall support maintain specification and parameters in equipment master data.	Customization

4.8.1.2.1	System shall suport calibration and maintenance plan locally.	Customization
4.8.1.2.2	System shall remind user when instrument maintenance expire in folder.	Configuration

3.2.1.6.3 Function Design

3.2.1.6.3.1 Design Overviews

When the set instrument maintenance time arrives, the system will automatically create a maintenance task and notify the maintenance owner by email.



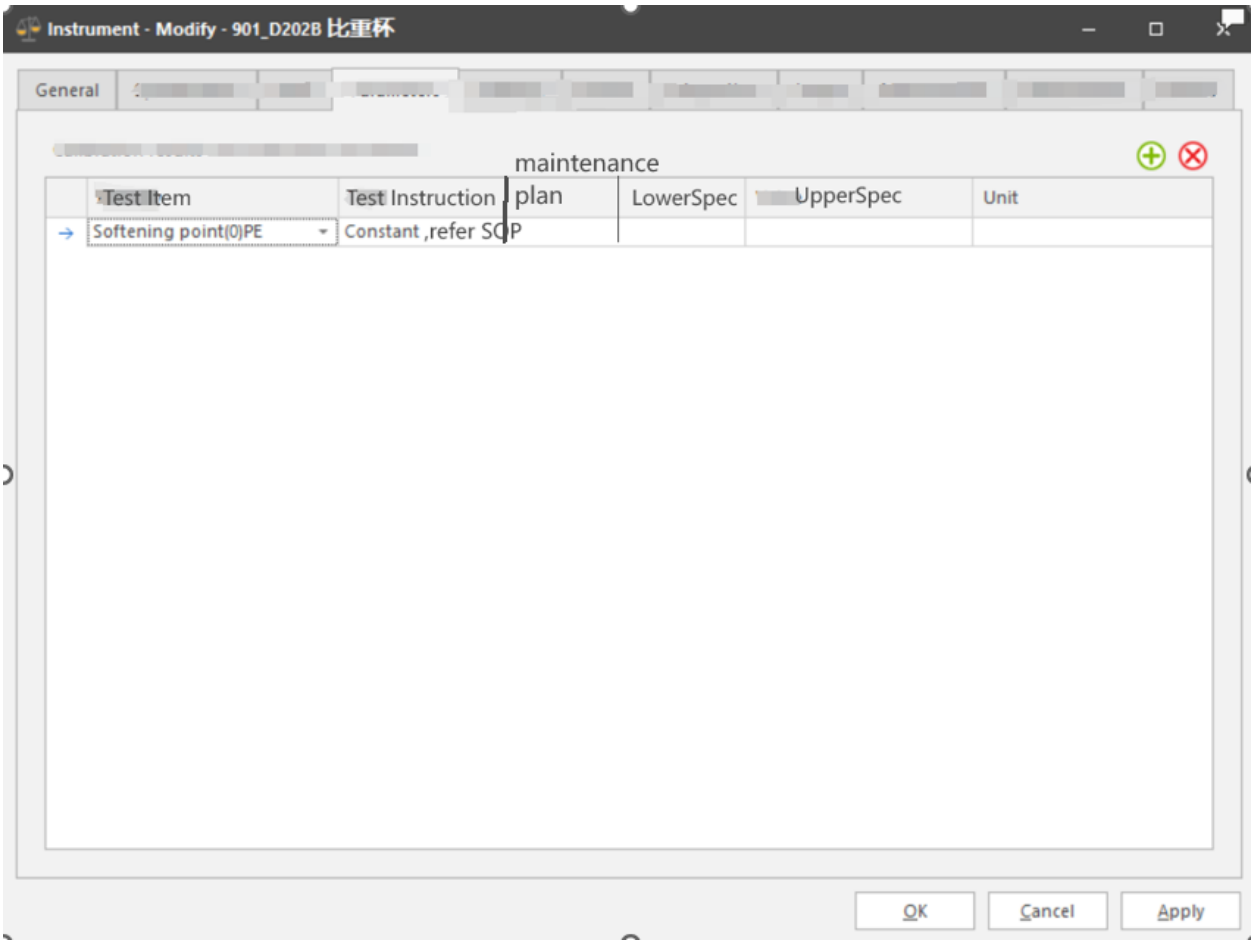
3.2.1.6.3.2 Process Design

Master Data

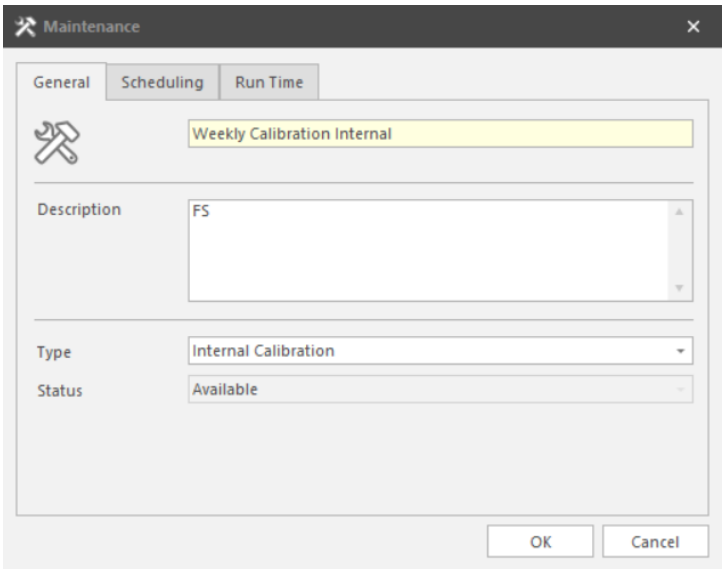
Test items:

In instrument master data, users can add test items in test data page, including maintenance name, test item name, test instructions, lower spec (optional), upper spec (optional) and unit (optional).

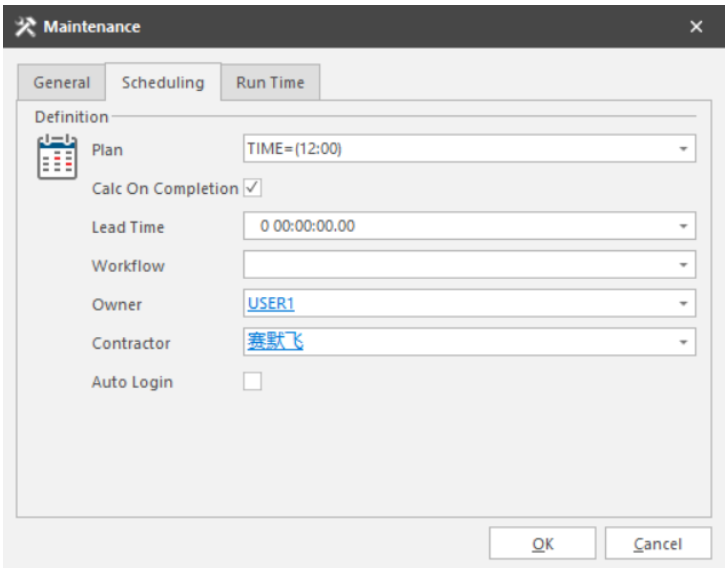
The test items will be used based on maintenance plan when executing calibration.



Calibration Plan:



Instrument Calibration plan is maintained in maintenance page of instrument. Add a maintenance task of internal calibration (or external calibration) type.



Add maintenance plan information on the scheduling page. Plan defines when the maintenance must be performed.

Lead time is the notice period for the maintenance schedule.

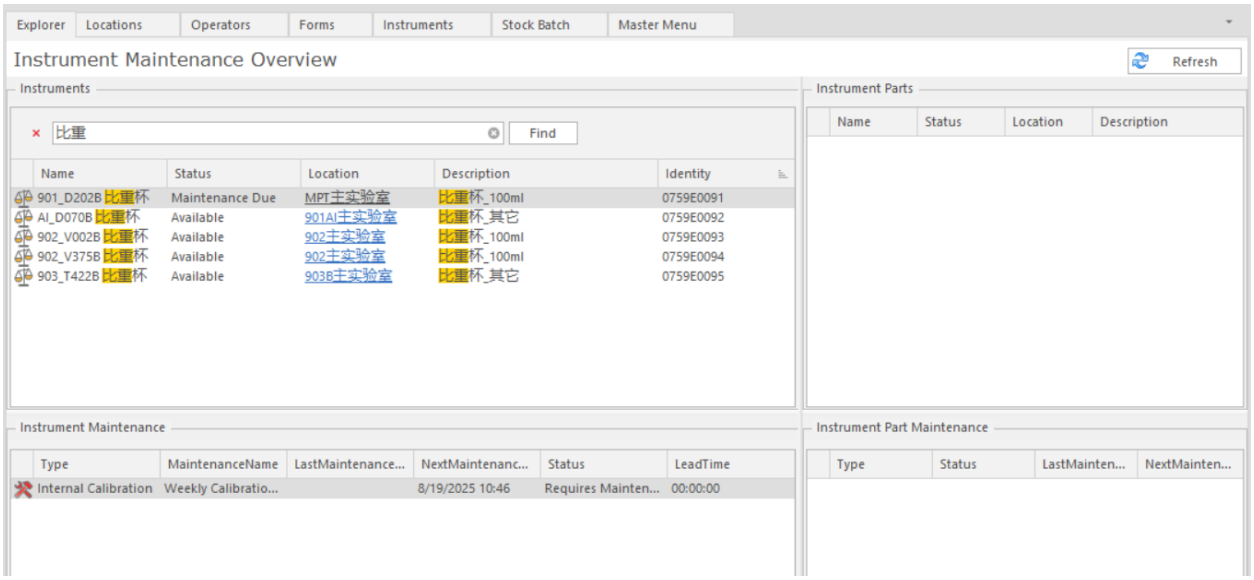
QCLAD will send the person specified in the Owner prompt an email to inform them when the maintenance task will become due.

The Lead Time defines the interval between QCLAD sending the email and the due date for the maintenance task.

Maintenance Tasks

The instrument maintenance schedule can be viewed by instruments in the instrument maintenance overview dashboard.

When the scheduled maintenance time arrives, the system will change the maintenance status to *requires maintenance*. The instrument status will become *maintenance due*.



Perform Calibration Internal

User can perform calibration by right click the instrument and click *Start Calibration Task* or right click on the maintenance item and click *Start Calibration Task*.

QCLAD will create a calibration task and pop up an instrument calibration window. Test information maintained in master data will be automatically carried over.

See the test item table below. Users can enter up to five results, and the system will automatically calculate the average value. Users can specify upper and lower limit values and manually select pass/fail.

Test item	Test instructions	Result1	Result2	Result3	Result4	Result5	Average	Lower specification limit	Upper specification limit	Pass/Fail	Comments
Vis1											
Vis2											
Vis3											

The system will automatically determine whether the test results pass or fail based on the average value and upper and lower limits when user enter the results.

When user click OK, calibration task is completed and the status of instrument will become available.

User can right click the available calibration task and click *upload calibration attachment* button to upload the attachment. QCLAD will give options allow user to choose if the calibration is pass or fail. When user click OK, QCLAD will pop up a window asking the user if they need to complete the calibration task. If click yes, calibration task is completed and the status of instrument will become available.

Change Status

User can change the instrument status to unavailable. Comments is mandatory when unavailable an instrument.User can change the instrument status to available. Comments is mandatory when available an instrument.

Perform Calibration External

User can find available calibration tasks in *Calibration Tasks* folder

User can right click the available calibration task and click *upload calibration attachment* button to upload the attachment. After uploading the attachment, QCLAD will pop up a window asking the user if they need to complete the calibration task. If click yes, calibration task is completed and the status of instrument will become available.

Calibration Report

QCLAD shall allow users to create calibration reports, and the reports should not contain blank test results.

3.2.2 Sample Test Assignment Procedure

3.2.2.1 Business Scenarios

For the task assignment function, it is desired to add the function of automatically assigning tasks to operators based on the phaseA manual assignment function. The system needs to automatically assign tasks based on shift scheduling information and operator qualifications, and it also needs to take into account the handling of shift handover situations.

3.2.2.1.1 RTM Mapping

URS Ref.	Requirement	Type
4.1.2.1.4	System shall be able to assign test based on the versatility matrix from OSDP system once integrated.	Customisation
4.2.1.1.12	System shall be able to assign test based on the versatility matrix from OSDP system once integrated.	Customisation
4.2.1.1.13	System shall be able to assign test based on the sample status (Waiting to start).	Customisation
4.2.1.1.14	System shall be able to auto assign for the current date based on the schedule configured.	Customisation
4.2.1.1.15	System shall be able to monitor samples on a certain interval basis and assign it.	Customisation
4.2.1.1.17	System shall be able to assign samples based on SAP test skip lot rules.	Customisation
4.2.1.1.18	System shall be able to assign samples based on other pending QC works.	Customisation
4.2.1.2.1	System shall be able to modify schedule.	Customisation
4.2.1.3.1	System shall be able to allow key user modify task based on urgency test.	Configuration
4.2.1.4.2	System shall be follow FIFO rules based on sample receiving date when assigning tasks.	Customisation
4.2.1.5.1	System shall assign task based on priority rules once sample arrives.	Customisation

4.2.1.6.1	System shall allow to create tasks based on calendar schedule.	Customisation
4.2.1.9.1	System shall check skill when assigning tasks to operator.	Customisation

3.2.2.2 Function Design

When a sample is received and the status changes from unavailable to available, it is considered that the tests under the sample can be assigned automatically and the system will assign the test according to the following rules.

3.2.2.2.1 Sample Assignment upon Receipt

When a sample is received, automatically assign its tests based on the following logic:

a. Find available personnel who are:
   - On duty that day
   - Match the Plant, WorkCenter (from personnel's WorkCenters), and Production Line (from personnel's Production Lines)

b. Check if the sample's tests require training courses:
   - If **yes**: Assign tests individually to personnel qualified for each test's training course.
     - **Exception**: If personnel meet condition (a) **and** are qualified for **all** tests in the sample, assign the entire sample to them.

c. If **no tests** require training courses: Assign all tests in the sample to one person.

d. When multiple tests under the same TestMethod in the same sample have the same eligible staff, these multiple tests will be assigned to the same person.

e. Multiple eligible personnel:
   - Assign to the person with the lightest workload (fewest unfinished samples with status "V").
   - If equal, assign to the first person in the list.

f. No eligible personnel found: Do not assign the test/sample.

g. Use the ‘Starter’ field in the test table to automatically determine whether the test is assigned based on whether it is empty.

3.2.2.2.2 Scheduled Background Assignment

Set up a scheduled task to assign unassigned tests with this logic:

a. Only assign unassigned tests.

b. Priority order:

   1. IPQC-type tests
   2. Truck-type tests
   3. Other types

c. Assign oldest received tests first.

d. Find personnel who are:

   - On duty that day
   - Match the Plant, WorkCenter (from personnel's WorkCenters), and Production Line (from personnel's Production Lines).

e. Training course check (same logic as 1.b).

f. If no tests require training: Assign entire sample to one person (same as 1.c).

g. When multiple tests under the same TestMethod in the same sample have the same eligible staff, these multiple tests will be assigned to the same person.

h. Multiple eligible personnel: Same as 1.d.

i. No eligible personnel: Do not assign.

j. Frequency: configured by the user. it is recommended to do this once a day.

3.2.2.2.3 Task Release during Shift Handover

When personnel end their shift:

- Release all their unfinished tasks (status "V", available).

  Behavior: Users can select multiple tests or samples assigned to themselves, right-click 'Task Release', and release the selected tests or samples assigned to themselves, i.e., set the 'Starter' field in the test table to empty.

  Note: In case the user did not release or forgot to release the task, the system will not process it in any way.

- These tasks become available for reassignment via the scheduled background assignment (Section 2).

- **Add"My Tasks" Folder**:

  Purpose: Streamline test management by allowing users to quickly locate and release their unfinished tests during shift changes.

Access: Tests are filtered by the logged-in user and can be configured by an administrator.

3.2.3 MES Interface

3.2.3.1 Introduction

This document describes the system functionality for the integration between QCLAD (SampleManager) and MES (Manufacturing Execution System) as part of Phase B implementation.

3.2.3.2 Purpose

The purpose of this document is to provide detailed specifications for the MES-QCLAD integration, covering JIT (Just-In-Time) and IPQC (In-Process Quality Control) processes, including interface requirements, data mapping, and system logic.

3.2.3.3 RTM Mapping

URS Ref.	Requirement	Type
4.1.1.2.2	System shall be able to trigger a resample from MES.	Customisation
4.1.1.17.3	System shall be automatically login samples from MES.	Customisation
4.2.1.9.2	System shall be able to support skill shall be sync with OSDP system if there is OSDP in plant. Refer to 4.5.1.14	Customisation
4.2.1.11.1	System shall create tasks for samples based on specific frequency.	Customisation
4.5.1.12	System shall push notify information(including resample and adjustment) to Henkel smart factory.	Interface
4.5.1.13	System shall pull result information from Henkel smart factory.	Interface
4.5.1.14	System shall support to upload result value from QCLAD to Henkel smart factory.	Interface
4.5.1.15	System shall support to upload sample states from QCLAD to Henkel smart factory.	Interface
4.5.1.16	System shall push/pull adjustment quantity information from/to Henkel smart factory.	Interface
4.5.1.17	System shall pull operator skill information from Henkel Skill Matrix.	Interface

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4.8.1.5.2	System shall sync user's testing skill from Henkel skill matrix tool.	Interface
4.8.2.3.2	System shall sync user's testing skill from Henkel skill matrix tool.	Interface

3.2.3.4 Scope

The scope of this document includes:

- JIT sample processing and result communication to MES
- IPQC sample processing and result communication to MES
- Production line information synchronization
- Estimate completion time information synchronization
- Interface specifications and data mapping

Note: *The JIT and IPQC interfaces are the same interface, the JSON format is the same, only the content of the return will be different.*

3.2.3.4.1 JIT Process Integration

3.2.3.4.1.1 Business Scenario

In the JIT process, samples are created manually in QCLAD before the inspection lot exists in SAP. After completing testing and authorization in QCLAD, the quality results must be sent to MES. When SAP later generates an inspection batch, QCLAD will match it to the existing sample.

3.2.3.4.1.2 Interface URL address

QCLAD will expose an API for MES to send JIT sample information and provide an API endpoint to receive results from QCLAD.

3.2.3.4.1.3 JIT Sample Info Receiving (QCLAD to MES)

1. Vegam will expose one API (path will be provided later), to which data will be send from QCLAD once QC is done for JIT sample number.
2. This API path and authentication credentials like client ID and client secret will be shared by Vegam. Using these credentials QCLAD need to get bearer token using Vegam oAuth API (path will be provided later).
3. QCLAD need to send sample QC data as below JSON with authentication as bearer token which is generated in step 2.

Example request JSON:

```
{
```

```
        "sampleType": "RECEIVING_SAMPLE",
        "sampleLabelNumber": "0000216316",
        "processOrder": null,
        "batchNumber": "A1233",
        "materialNumber": "383743",
        "materialDescription": "Adeka Hardener EH 4070 S",
        "plantCode": "0799",
        "activityNumber": null,
        "sampleApprovalRejectedTime": "YYYY-MM-DDTHH:mm:ssZ",
        "operatorName": "4454",  // To be discuused
        "QCSampleResult": "pass", //Pass or Reject
        "IsQCAdjustment": false,
        "IsFinalRejection": false,
        "adjustMaterials": [],
        "consesssionRelease": false,
        "resampleRequired": false,
        "remark": "",
        "QCLabItemResultList ":[]
    }
```


Example response JSON:


```
{
    "MesSampleId": "0000574979",
    "Status": 0,
    "Message": ""
}
```


3.2.3.4.1.4 Error handling

ReturnCode	Description	Action By QCLAD
0001	Success	No Action
0002	API Authentication Error	Correct Authentication and Resend
0003	Invalid Sample Number	Correct sample number and Resend

0004	Invalid date time format	Correct the data and Resend
0005	Invalid JSON	MES Will take care further process

3.2.3.4.1.5 Field Mapping

No.	MES Field	Description	QCLAD Field Mapping
1.	sampleType	sample Type	sample.HenInspectionCategory
2.	sampleLabelNumber	Unique ID for the sample	sample.HenMesId
3.	processOrder	SAP Process Order	Sample.HenProcessOrder
4.	batchNumber	SAP Batch Number	sample.BatchName
5.	materialNumber	SAP Material Code	sample.Substance.Identity
6.	materialDescription	Description of material	sample.Substance.Name
7.	plantCode	SAP Plant	sample.HenPlant
8.	activityNumber	SAP phase in the routing	sample.HenPhraseNumber
9.	sampleApprovalRejectedTime	Date and time sample was approved/rejected	sample.DateAuthorised
10.	operatorName	Operator who approved sample	sample.Authoriser
11.	QCSampleResult	Approved or Rejected	sample.Status and sample.OnSpec
12.	isQCAdjustment	is QC Adjustment	NA
13.	adjustMaterials	The material information that needs to be adjusted can be zero or more.	NA

14.	consesssionRelease	consesssion Release	NA
15.	resampleRequired	If the phase required re sample	NA
16.	remark	Comments	NA
17.	QCLabItemResultList		NA

3.2.3.4.1.6 System Logic

1. When a JIT sample is authorized in QCLAD, the system sets hen_MES_to_be_sent to true
2. A background task (HenSendJITResultToMESTask) runs every 5 minutes
3. The task retrieves all JIT samples with hen_MES_to_be_sent=true and hen_MES_Has_sent=false
4. For each sample, QCLAD calls the MES interface to send sample information.
5. After successful transmission, hen_MES_Has_sent is set to true
6. In QCLAD interface log will have separate MES interface log for investigations.

3.2.3.4.2 IPQC Process Integration

3.2.3.4.2.1 Business Scenario

For IPQC samples, QCLAD will send test results to MES after sample authorization. This includes passing results, failed results with adjustment requirements, or resample requests.

3.2.3.4.2.2 Interface URL address

QCLAD will expose an API for MES to send IPQC sample information and provide an API endpoint to receive results from QCLAD.

3.2.3.4.2.3 Interface requirements

- After completion of IPQC testing, QCLAD will push the results with usage decision, MIC results, MES Sample ID and adjustment material, quantity (in case there is an adjustment)
- In QCLAD, there should be an option to select the usage to be sent to MES
  - In case all MICs are passed, QCLAD will push “Accept” code.
  - In case one or MICs fail, QCLAD should need to send the following based on QC selection
    - ❖ Rejected with adjustment. Need to add a flag whether it is QC adjustment or manufacturing adjustment. If it is QC adjustment, will

populate adjustment material, adjustment quantity, adjustment quantity UOM, storage location and free text with comments of QC. MES has to create a new adjustment phase with all details provided by QC. In case there is no stock, MES has to wait in this phase till stock arrives and then continue to adjustment phase.

If it is not done by QC, need to get the details of adjustment from manufacturing.

- Rejected with resample. QCLAD need to give an option to write instruction note for this scenario. MES need to send a new sample without any adjustment. QCLAD will provide a popup message for user to resample. The user in QCLAD create resample for virtual sample id and wait for MES label to receive it.
- Rejected as Off-spec. On receiving this, MES has to discontinue from going to next phase or if it is last phase should not go with packaging

Accepted with Concession- On receiving this MES has to continue the next phase. If it is last phase should continue with packaging.

3.2.3.4.2.4 IPQC Sample Info Receiving (QCLAD to MES)

1. Vegam/DCS will expose one API (path will be provided later, same IPQC info receiving API will be used), to which data will be send from QCLAD once QC is done for IPQC sample number.
2. This API path and authentication credentials like client ID and client secret will be shared by Vegam (these will be same as IPQC receiving info).
   Using these credentials QCLAD need to get bearer token using Vegam oAuth API (path will be provided later).
3. QCLAD need to send sample QC data as below JSON with authentication as bearer token which is generated in step 2.

Example request JSON:


```
{
  "SampleType": "IPQC",
  "SampleLabelNumber": "371584",
  "ProcessOrder": "001004172754",
  "BatchNumber": "0007447212",
  "MaterialNumber": "1392015",
  "MaterialDescription": "Liquid Detergent Lavender",
  "PlantCode": "0799",
  "ActivityNumber": "0023",
  "AampleApprovalRejectedTime": "YYYY-MM-DD HH:mm:ss",  //plant time zone
  "OperatorName": "John Doe",
  "QCSampleResult": "pass", //Pass or Reject
  "IsQCAdjustment": true,  [True/False]
  "IsFinalRejection": false,
```

```
"AdjustMaterials": [
  {
    "MaterialNumber": "123456",
    "BatchNumber": "BT01XZSH",
    "Quantity": 10,
    "UOM": "KG",
    "StorageLocation" : "ST01"
  }
],
"ConsesssionRelease": false, //true or false
"ResampleRequired": false, //true or false
"Remark": "",// commet based on sample level
"QCLabItemResultList":[{
    "InspectionLot":"1234567890",
    "InspectionCharNumber":"0020",
    "MasterInspectionCharacteristics":"00000069",
    "InspectionCharacteristicsText":"TENSILE LAP-SHEAR STRENGTH",
    "InspectionMethod":"00595101",
    "InspectionMethodText":"Shear Strength-Block Specimens",
    "LowerSpecLimit":254.5,
    "UpperSpecLimit":290.6,
    "DecimalPlace":1,
    "MeasureUnit":"N/mm2",
    "ResultRecordType":"N",  //can be N-Numeric or O-Options
    "Result":"275.7", //if result is type is N then decimal result as string else pass/fail
    "Comments":""
  }
]
 }
```


Example response JSON:


```
{
    "MesSampleId": "0000574979",
    "Status": 0,
    "Message": ""
}
```


3.2.3.4.2.5 Error handling

ReturnCode	Description	Action By QCLAD

0001	Success	No Action
0002	API Authentication Error	Correct Authentication and Resend
0003	Invalid Sample Number	Correct sample number and Resend
0004	Invalid date time format	Correct the data and Resend
0005	Invalid JSON	MES Will take care further process

3.2.3.4.2.6 Field Mapping

No.	MES Field	Description	QCLAD Field Mapping
1.	sampleType	sample Type	sample.HenInspectionCategory
2.	sampleLabelNumber	Unique ID for the sample	sample.HenMesId
3.	processOrder	SAP Process Order	Sample.HenProcessOrder
4.	batchNumber	SAP Batch Number	sample.BatchName
5.	materialNumber	SAP Material Code	sample.Substance.Identity
6.	materialDescription	Description of material	sample.Substance.Name
7.	plantCode	SAP Plant	sample.HenPlant
8.	activityNumber	SAP phase in the routing	sample.HenPhraseNumber
9.	sampleApprovalRejectedTime	Date and time sample was approved/rejected	sample.DateAuthorised
10.	operatorName	Operator who approved sample	sample.Authoriser
11.	QCSampleResult	Approved or Rejected	sample.Status and sample.OnSpec

12.	isQCAdjustment	is QC Adjustment	NA
13.	adjustMaterials	The material information that needs to be adjusted can be zero or more.	NA
14.	materialNumber	Material used for any adjustment during sampling	NA
15.	batchNumber	bacth number of adjustment material	NA
16.	quantity	Amount of material sampled or adjusted	NA
17.	uom	unit	NA
18.	storageLocation	storage Location	NA
19.	consesssionRelease	consesssion Release	NA
20.	resampleRequired	If the phase required re sample	NA
21.	remark	Comments	NA
22.	QCLabItemResultList		NA
23.	InspectionLot	Lot number for IPQC	Hen_inspectionlot
24.	inspectionCharNumber	Order no	test.hen_orderno
25.	masterInspectionCharacteristics	MIC_code	test.hen_mic_code
26.	inspectionCharacteristicsText	short_text	test.hen_short_text
27.	inspectionMethod	Test Method	Test.TestMethod
28.	inspectionMethodText	Test Method Name	Test.analysis.name
29.	lowerSpecLimit	lower Spec Limit	MLP_VALUES.MIN_LIMIT

30.	upperSpecLimit	upper Spec Limit	MLP_VALUES.MAX_LIMIT
31.	decimalPlace	PLACES	result.places
32.	measureUnit	unit	result.Units
33.	resultRecordType	result Type	result.resultType
34.	InspectionResult	result value	result_raw_result
35.	codeText	OnSpec	result.hen_on_spec

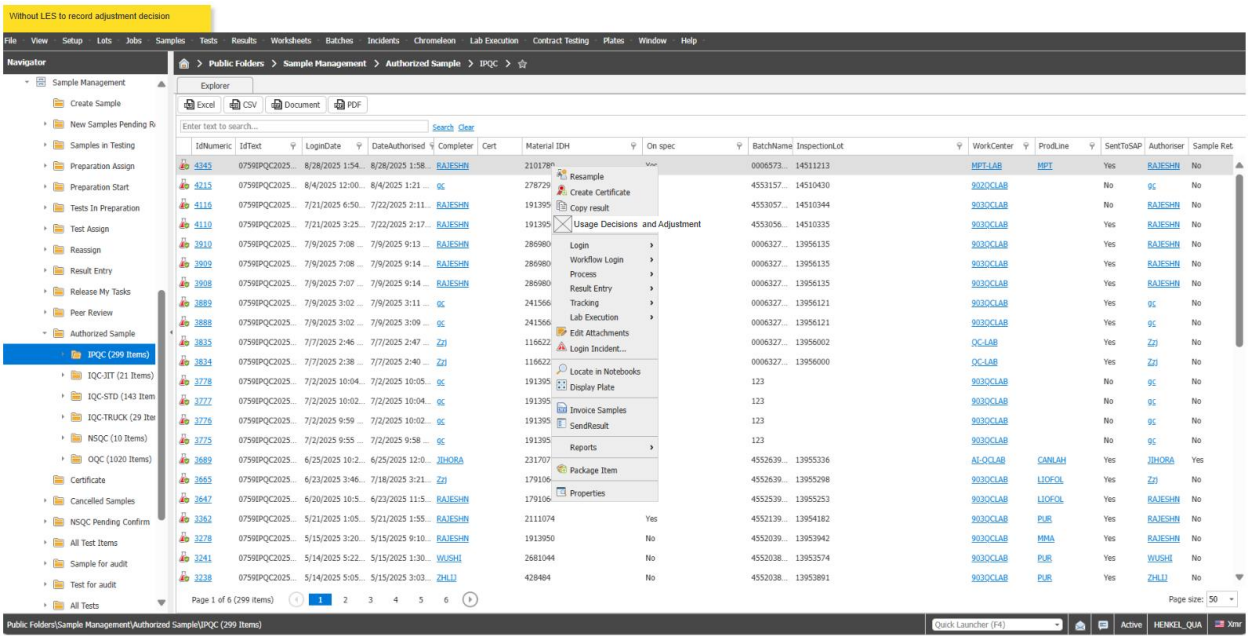
3.2.3.4.2.7 System Logic

1. A background task (HenSendResultToMESTask) runs every 5 minutes
2. The task retrieves all IPQC samples with hen_MES_to_be_sent=true and hen_MES_Has_sent=false
3. For each sample:
   - If passed: Send decision "Accept" without adjustment information
   - If failed: Four options as below:
     1. Send decision "Accept with concession " without adjustment information
     2. Send decision "Reject with adjustment" with adjustment information
     3. Send decision "Reject with resample" without adjustment information
     4. Send decision "Reject with off spec" without adjustment information

Details: Refer to prototype as below:

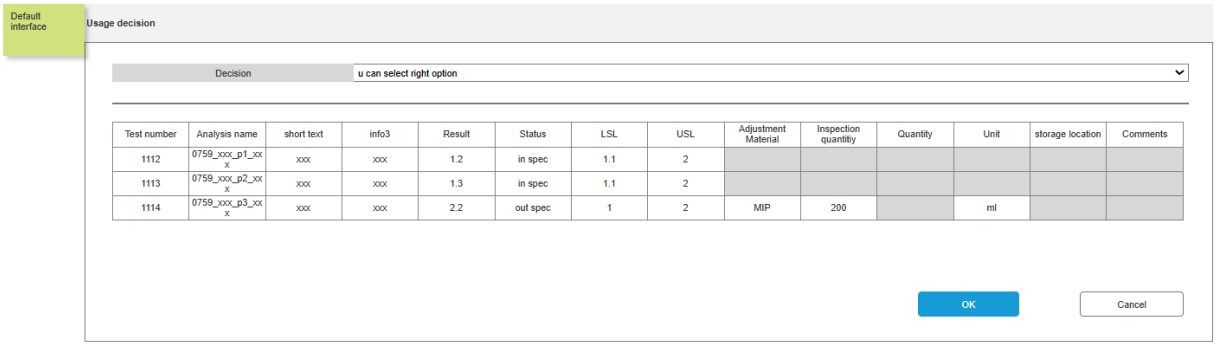
Path: Sample management cabinet→Authorized sample folder→IPQC subfolder.

Under authorized sample folder, only for IPQC sample with “usage done” is ‘No’ will highlight in pink to reminder user the sample without decisions.



Notes: the “Usage decision and adjustment” button (button name can be customized by user if needed) will show up when the sample type is ‘IPQC’ and Usage decision filed is ‘No’.

Step1: right click “Usage Decisions and adjustment” Master Menu on test item. Will pop up default windows as below:



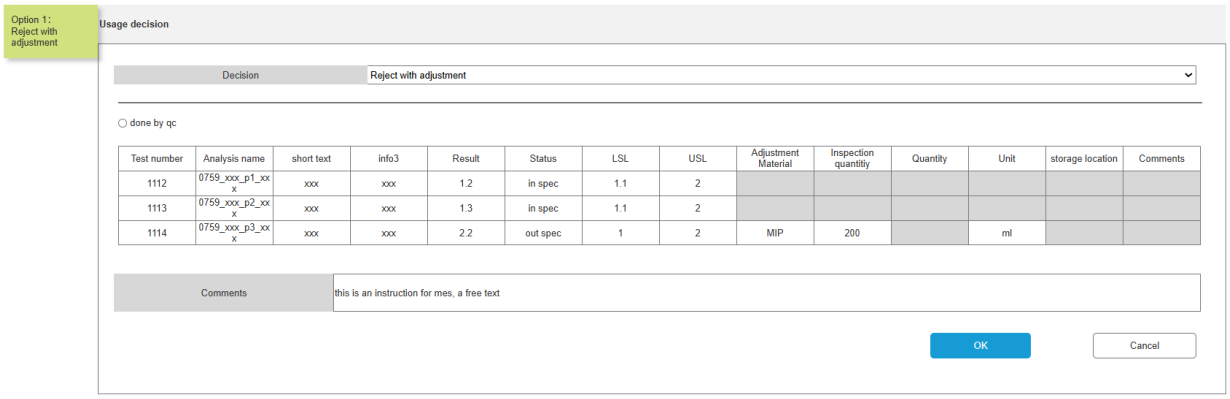
Test number	Analysis name	short text	info3	Result	Status	LSL	USL	Adjustment Material	Inspection quantity	Quantity	Unit	storage location	Comments
1112	0759_xxx_p1_xxx	xxx	xxx	1.2	in spec	1.1	2						
1113	0759_xxx_p2_xxx	xxx	xxx	1.3	in spec	1.1	2						
1114	0759_xxx_p3_xxx	xxx	xxx	2.2	out spec	1	2	MIP	200		ml		

Note: This test item information form is for display purposes only. Users cannot modify it; they may only select “Reject with adjustment” by the options.

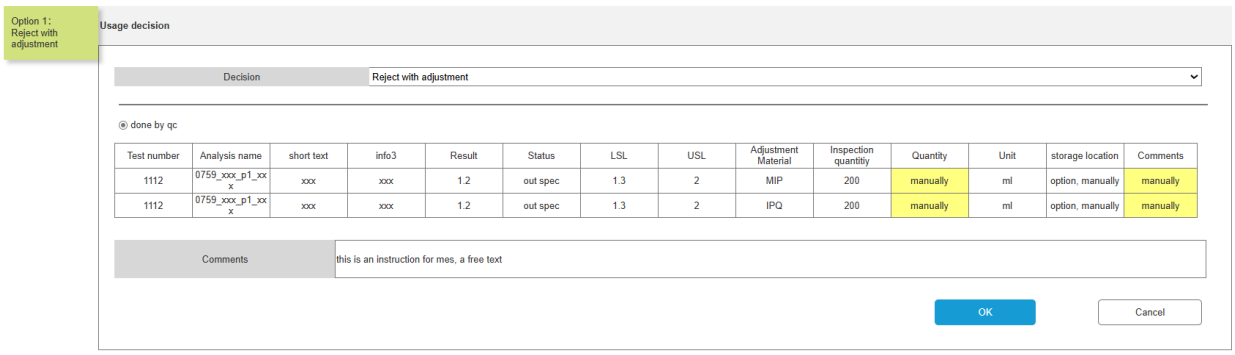
This form will be shown under all options for the user's reference, except for 'Reject with Adjustment' and Click 'Done by QC', as this option will only be shown for out-of-spec test items.

User select each option will show up as below:

Option1: user select “reject with adjustment” without clicking “done by qc” as below:



Option1: user select “reject with adjustment” with clicking “done by qc” as below:



Users can input or select values for the three fields: “Quantity,” “Storage Location,” and “Comments.”

Fields description:

Field Name	**Example Value**	**Mandatory?**	**Input Method**
Test number	1112	Yes	Auto
Analysis name	0759_xxx_p1_xxx	Yes	Auto
short text	xxx	Yes	Auto
info3	xxx	No	Auto
Result	1.2	Yes	Auto
Status	out spec	Yes	Auto

Field Name	Example Value	Mandatory?	Input Method
LSL	1.3	Yes	Auto
USL	2	Yes	Auto
Adjustment Material	MIP	Yes	Auto capture from adjustment table
Inspection quantity	200	Yes	Auto
Quantity	manually	Yes	Manual
Unit	ml	Yes	Auto
storage location	option, manually	No	Auto capture from adjustment table
Comments	manually	Yes	Manual

These fields are all captured from master data “Adjustment table”.

Option 2: Reject with resample

Usage decision

Decision: Reject with resample

Please Resample!
User need to resample immediately in case the mes label already send to QCLAD but no sample created!

Test number	Analysis name	short text	info3	Result	Status	LSL	USL	Adjustment Material	Inspection quantitiy	Quantity	Unit	storage location	Comments
1112	0759_xxx_p1_xxx	xxx	xxx	1.2	in spec	1.1	2						
1113	0759_xxx_p2_xxx	xxx	xxx	1.3	in spec	1.1	2						
1114	0759_xxx_p3_xxx	xxx	xxx	2.2	out spec	1	2	MIP	200		ml		

Comments: this is an instruction for mes, a free text

OK Cancel

Option 3: Reject with off sepc

Usage decision

Decision: Reject with off spec

Test number	Analysis name	short text	info3	Result	Status	LSL	USL	Adjustment Material	Inspection quantitiy	Quantity	Unit	storage location	Comments
1112	0759_xxx_p1_xxx	xxx	xxx	1.2	out spec	1.3	2	IMP	200		ml		
1113	0759_xxx_p2_xxx	xxx	xxx	1.3	out spec	1.4	2	DING	200		ml		
1114	0759_xxx_p3_xxx	xxx	xxx	2.2	out spec	1	2	MIP	200		ml		

Comments: this is an instruction for mes, a free text

OK Cancel

Option 4: Accept with concession

Usage decision

Decision: Accept with consession

Test number	Analysis name	short text	info3	Result	Status	LSL	USL	Adjustment Material	Inspection quantitiy	Quantity	Unit	storage location	Comments
1112	0759_xxx_p1_xxx	xxx	xxx	1.2	in spec	1.1	2						
1113	0759_xxx_p2_xxx	xxx	xxx	1.3	in spec	1.1	2						
1114	0759_xxx_p3_xxx	xxx	xxx	2.2	out spec	1	2	MIP	200		ml		

Comments: this is an instruction for mes, a free text

OK Cancel

Option 5: Accept

Usage decision

Decision: Accept

Test number	Analysis name	short text	info3	Result	Status	LSL	USL	Adjustment Material	Inspection quantitiy	Quantity	Unit	storage location	Comments
1112	0759_xxx_p1_xxx	xxx	xxx	1.2	in spec	1.1	2						
1113	0759_xxx_p2_xxx	xxx	xxx	1.3	in spec	1.1	2						
1114	0759_xxx_p3_xxx	xxx	xxx	0.9	out spec	1	2						

Comments: this is an instruction for mes, a free text

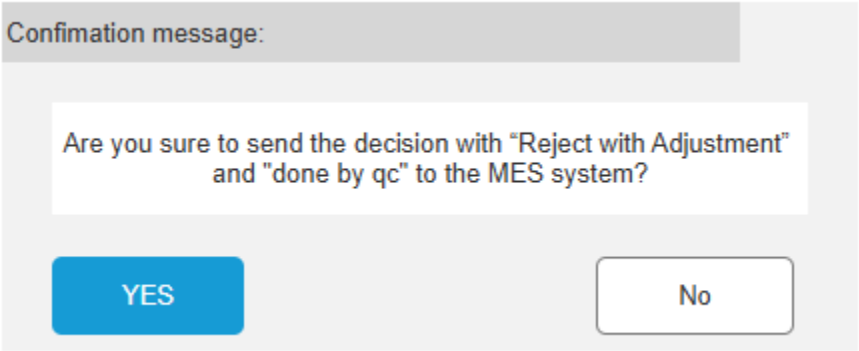
OK Cancel

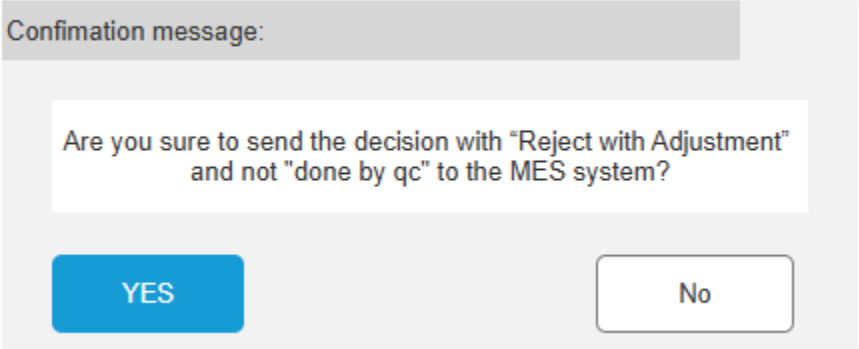
Step2: If users select the usage decision and enter the mandatory information, then click OK button. Will pop up confirmation check window as below.

If users click Cancel button, the "Usage decision"window will cancel and the inforamtion won't store.

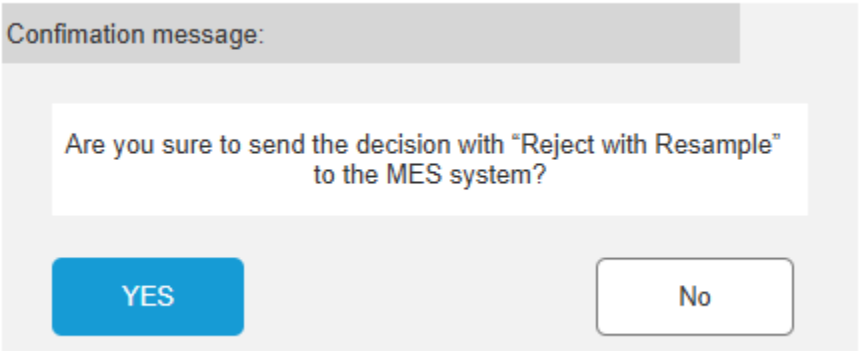
For option 1: "Reject with Adjustemnt"

will have two window as different selection on "done by qc"

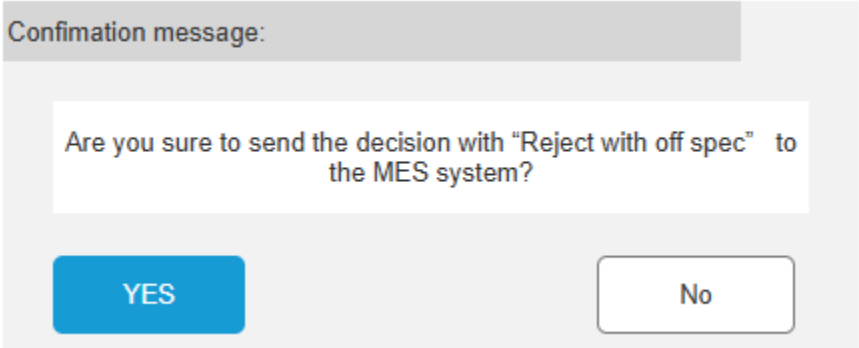




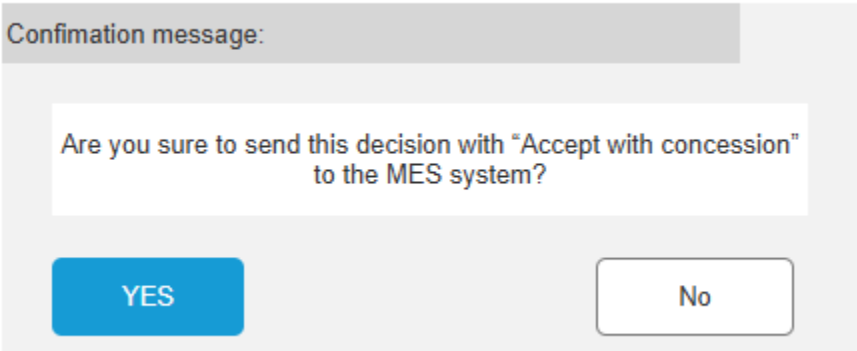
For option 2: "Reject with Resample"



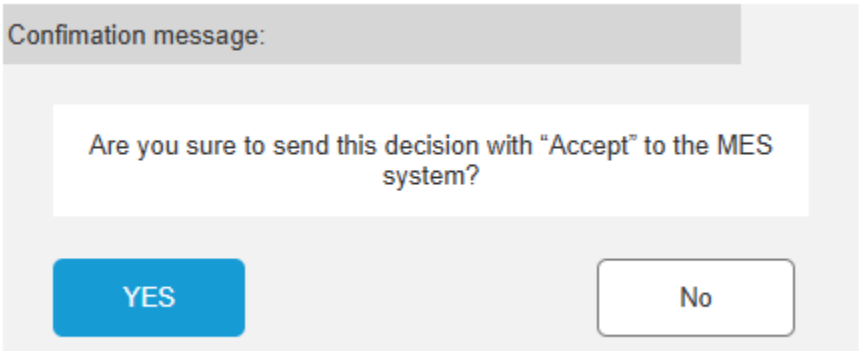
For option 3: "Reject with off spec"



For option 4: "Accept with concession"



For option 5: "Accept"



Step3: If users click

button. The "Usage done"field will change to YES and Hen_to_be_sent is set to TRUE. Meantime higlight in pink will disapper.

If users click

button. The fields and highlight won't change.

4. Once the user has finished making their usage decisions, the “usage done” field on sample table will change to 'Yes' and Hen_to_be_sent is set to TRUE, the adjustment information will be stored into the json file.
5. Add a new field to the sample table to indicate which system the sample was sent to, enabling the plant team to check whether the results have been pushed to the MES or DCS.
6. After successful transmission, Hen_MES_Has_sent is set to TRUE.
7. Adjustment master data maintenance and information entry
   Add a new adjustment table: Hen_adjustment_table:
   Example:

MLP	IDH	Analysis	Analysis Name	TestMethod	Above_standard	Adjustment_Material	storage location	Plant	WorkCenter	Product Line	modify_on	modify_by	Group
06-1272-1	06-1272	TM001	SOD_MIC01	LOMSOD01	Above	DNIP	0005	0759			user1	2025-06-18 11:40	0759
06-1272-1	06-1272	TM001	SOD_MIC01	LOMSOD01	Above	YD128	0005	0759			user1	2025-06-18 11:40	0759
06-1272-2	06-1272	TM001	SOD_MIC01	LOMSOD01	Above	S3	0005	0759			user1	2025-06-18 11:40	0759
06-1272-2	06-1272	TM001	SOD_MIC01	LOMSOD01	Above	S4	0005	0759			user1	2025-06-18 11:40	0759
06-1272	06-1273	TM001	SOD_MIC01	LOMSOD01	Below	M5	0005	0759			user1	2025-06-18 11:40	0759
06-1272	06-1274	TM002	剪切强度_MIC02	LOMXX02	Below	CG 1200	0005	0759			user1	2025-06-18 11:40	0759
06-1272	06-1275	TM003	剪切强度_MIC02	LOMXX02	Below	UR 300	0005	0759			user1	2025-06-18 11:40	0759

   Fields description:

Field	Is it a primary key	Field type	Mandatory	Description
MLP	Y	Links Mlp_header table	Y	
IDH		Text		sample IDH Automatically imported from MLP
Analysis	Y	Links versioned_Analysis table	Y	
Analysis Name		text		Automatically imported from Analysis
TestMetthod		text		Automatically imported from Analysis
Above_standard	Y	Links Phrase	Y	Above standard or below standard Phrase options ：Above、Below
Adjustment_Material	Y	Links substance table	Y	adjustment IDH
Plant		Links hen_Plant table	Y	
WorkCenter		Links hen_WorkCenter table		
storage location		Links hen_Plant_location table		
Product Line		Links hen_Product_line table		
modify_on		Date		Automatically entered by the system
modify_by		Links personnel table		Automatically entered by the system
Group		Links group_header table		Automatically import Plant's Group

8. Production line master data configuration:
   - Configure whether IPQC needs feedback
   - Configure if need to send to MES or DCS.
9. Plant master data configuration:
   - Configure MES and DCS interface addresses

   Note: *The IPQC needs to be passed back to both the MES and the DCS. However, the JSON format of the request and response for the MES and DCS interfaces remains the same. The MES is*

authenticated by a token, while the DCS interface uses basic authentication with a username and password.

10. Plant Storage Location data distribution via SAP plant interface and storage
11. Add batch quantity and unit fields to SAP Inspection lot interface
12. JIT & IPQC result feedback interface calls (requires MES interface to first provide authentication and interface information)
    Sub-tasks:
    - Adjustment information entry
    - JIT & IPQC result feedback interface calls (requires MES interface to first provide authentication and interface information)
13. The system will give a pop-up with "Yes" or "No" to warn users whether you really need to do this or not. This is to prevent the accidental and this can create issues in supply chain.
14. The system will only give one decision to MES based on sample level.

3.2.3.4.3 Production Line Information Synchronization

3.2.3.4.3.1 Business Scenario

In QCLAD, the production line information needs to be synchronized with MES data. QCLAD will request this information from MES Label.

3.2.3.4.3.2 Requirements

- MES label will provide the production line information, QCLAD will capture the information and store in substance level. If no change, QCLAD will keep it as it is.

3.2.3.4.3.3 System Logic

1. For each MES label with production line information, QCLAD will retrieve the information and update to substance template.

3.2.3.4.4 API detail

3.2.3.4.4.1 API Authentication Mechanism

All APIs will use Owin OAuth 2.0 for Authentication. We are generating access token using the OAuth2.0 framework.

Access Token Request

When the user authorizes the application, the access token request will be sent to the authorization module within the server. The module will generate an access token for the client which is only valid for 30 minutes. All APIs are to be accessed using this access token

#	API Name	Description	Hosted On	Type
1	GetAccessToken	Get access token for accessing the required API	SmartFactory Solution	Pull

The API specifications is elaborated in the sections below.

GetAccessToken

The API provides access token for accessing SmartFactory Solution API. The API will be called by Rohlig before accessing any API of SmartFactory Solution. The API returns access token information.

Web service URL to access test server:
https://sfs-ap-test.henkelgroup.net/VegamOauthServer/GetAccessToken

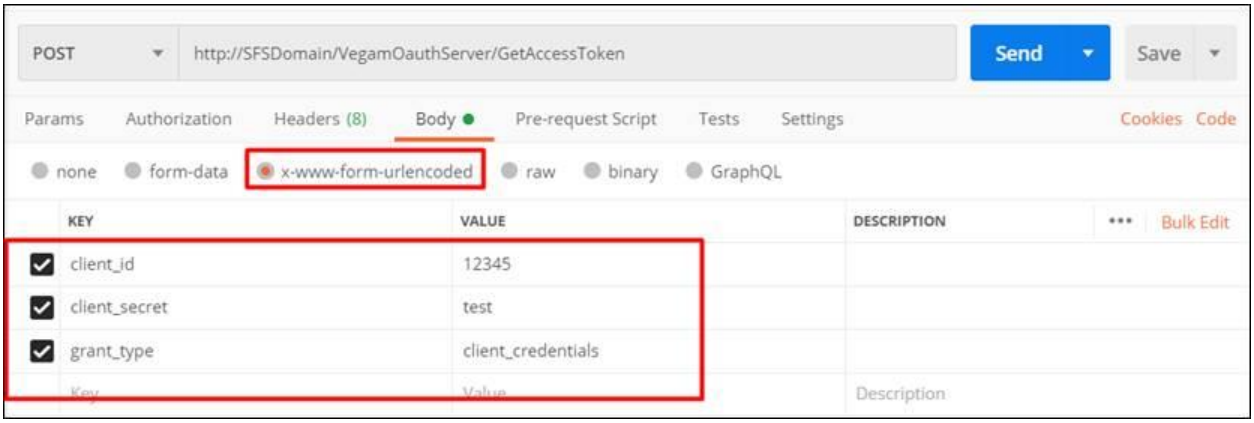
Web service URL to access production server:
https://ap-hipas.henkelgroup.net/VegamOauthServer/GetAccessToken

Web service Type: POST

Web service input parameters:

Parameter Name	Description	DateType	Is Mandatory
client_id	Required for access token generation. Details will be provided by SmartFactory solution.	VARCHAR	Yes
client_secret	Required for access token generation. Details will be provided by SmartFactory solution.	VARCHAR	Yes
grant_type	Required for access token generation. Must be set to the value **client_credentials**	VARCHAR	Yes

By default, these parameters must be **x-www-form-urlencoded** and specified in the request body (as shown in the sample below)



Web service output:

Output will be JSON response with access token as one of the parameters which need to be used for accessing AMS solution API.


```
{
    "access_token":
        "LR2ziWKVAu9XG3OrmOA4mOVnYAW9zmAB1seMDZ1iMippDnYN08wY5Ty4S4V8X1aXifqP6Et7kq_iHQKoLg8NDiuHf_9QVIJI_k5fUDBg_WQBntU
        ABE_oijsaqMDZZ4suViUsidC42KOKZ_hcfia2f57g67osCHZRhp8zjJLFX4omu5tj8OiJJ_MLaACnaJ1TlquldySG7NbAE0XKAzL27oEuFd3upVc
        ihCqahvpKzNc0UDI25RxgRBdtF7zF5hTmJ9nNpUMhYyOycOsbu4hSQydqL0jxIIKXMyRtUaZURE-vmgYSCLNE3jmZZyaiWa5Qch2cBCd2kuHJkQf
        tAcO2vQ",
    "token_type": "bearer",
    "expires_in": 599
}
```


In case of error below information with ErrorCode will be displayed

ErrorCode Information:

Error Code	Description
0001	Invalid client info
0002	Error while processing request


```
{
  "error": "",
  "error_description": ""
}
```



```
1 {
2     "error": "0001",
3     "error_description": "Invalid client info. Please contact administrator."
4 }
```


3.2.3.4.4.2 API Specifications

API is required to be developed to successfully integrate QCLAD with Vegam SmartFactory solution. API will need to be developed and exposed by Vegam SmartFactory solution, as explained in the table below.

#	API Name	Description	Hosted On	Type
2	UpdateSampleQualityInfo	Insert / Update Sample label quality details into SmartFactory solution	Vegam Domain	Push

The API specifications is elaborated in the below sections.

UpdateSampleQualityInfo

API will accept agreed JSON structure, details are as below.

Web service URL to access vegam test server:
https://sfs-ap-test.henkelgroup.net/ VegamOauthServer/api/data/UpdateSampleQualityInfo

Web service URL to access vegam production server:
https://ap-hipas.henkelgroup.net/ VegamOauthServer/api/data/UpdateSampleQualityInfo

Web service Type: POST

Web service input parameters for IPQC:

```
{
    "SampleType": "IPQC",
    "SampleLabelNumber": "371584",
    "ProcessOrder": "001004172754",
    "BatchNumber": "0007447212",
    "MaterialNumber": "1392015",
    "MaterialDescription": "Liquid Detergent Lavender",
    "PlantCode": "0799",
    "ActivityNumber": "0023",
    "AampleApprovalRejectedTime": "YYYY-MM-DD HH:mm:ss",  //plant time zone
    "OperatorName": "John Doe",
    "QCSampleResult": "Pass", //Pass or Reject
    "AdjustMaterials": [
        {
            "MaterialNumber": "123456",
            "BatchNumber": "BT01XZSH",
            "Quantity": 10,
            "UOM": "KG",
            "StorageLocation" : "ST01"
        }
    ],
    "ConsesssionRelease": false, //true or false
    "ResampleRequired": false, //true or false
    "Remark": "",
    "QCLabItemResultList":[{
            "InspectionLot":"1234567890",
            “InspectionCharNumber":"0020",
            "MasterInspectionCharacteristics":"00000069",
            "InspectionCharacteristicsText":"TENSILE LAP-SHEAR STRENGTH",
            "InspectionMethod":"00595101",
            "InspectionMethodText":"Shear Strength-Block Specimens",
            "LowerSpecLimit":254.5,
```

```
            "UpperSpecLimit":290.6,
            "DecimalPlace":1,
            "MeasureUnit":"N/mm2",
            "ResultRecordType":"N",  //can be N-Numeric or O-Options
            "Result":"275.7", //if result is type is N then decimal result as string else pass/fail
            "Comments":""
        }
    ]
}
```

Web service input parameters for JIT:

```
{
    "SampleType": "JIT",
    "SampleLabelNumber": "0000216316",
    "BatchNumber": "A1233",
    "MaterialNumber": "383743",
    "MaterialDescription": "Adeka Hardener EH 4070 S",
    "PlantCode": "0799",
    "SampleApprovalRejectedTime": "YYYY-MM-DD HH:mm:ss",  //plant time zone
    "OperatorName": "John Doe",
    "sampleStatus": "Approved", //Approved or Rejected
    "remark": ""
}
```

Above parameter should be passed in JSON format.

Web service output:

API will return an object as return details.

Return_Info Object Details:

Parameter Name	Description	DateType
ReturnCode		Varchar
Description		Varchar

Return Code Info

Return Code	Description	Action By QCLAD
0001	Success	No Action
0002	API Authentication Error	Correct Authentication and Resend
0003	Invalid Sample Number	Correct sample number and Resend

0004	Invalid date time format	Correct the data and Resend
0005	Invalid JSON	MES Will take care further process
0006	Estimated Completion time is passed	MES Will take care further process

Note: In the event of any network failure, QCLAD should resend the data to ensure successful processing.

3.2.4 DCS Interface

3.2.4.1 Introduction

This document describes the system functionality for the integration between QCLAD (SampleManager) and DCS as part of Phase B implementation.

3.2.4.2 Purpose

The purpose of this document is to provide detailed specifications for the DCS-QCLAD integration, covering IPQC (In-Process Quality Control) processes, including interface requirements, data mapping, and system logic.

3.2.4.3 Requirements

Key Points about DCS and MES Integration:

1. **Parallel Systems**:
   DCS requires the same callback mechanism from QCLAD as MES, serving as two independent destination systems.
2. **Data Consistency**:
   Both systems share identical JSON formats and response structures for quality results.
3. **Key Differences**:
   - **Endpoint Configuration**: Unique URLs configured separately at **Plant level**
   - **Response Isolation**: Callback responses are system-specific (MES/DCS responses never intersect)

3.2.4.4 Configuration Logic

- Plant-Level: Configure MES/DCS URLs in Interface Config table
- ProductionLine-Level: Enable/Disable callback per system
- Transmission Rules:

- If both URLs configured → Send to both systems in parallel
- If only one URL configured → Send only to that system

3.2.4.5 Scope

The scope of this document includes:

- IPQC sample processing and result communication to DCS
- Interface specifications and data mapping

Note: *The JIT and IPQC interfaces are the same interface, the JSON format is the same, only the content of the return will be different.*

3.2.4.5.1 ***IPQC Process Integration***

3.2.4.5.1.1 Business Scenario

For IPQC samples, QCLAD will send test results to DCS after sample authorization. This includes passing results, failed results with adjustment requirements, or resample requests.

3.2.4.5.1.2 Interface URL address

QCLAD will expose an API for DCS to send IPQC sample information and provide an API endpoint to receive results from QCLAD.

3.2.4.5.1.3 Interface requirements

The same as MES interface, please refer to 3.2.2.4.2.3 Interface Requirement.

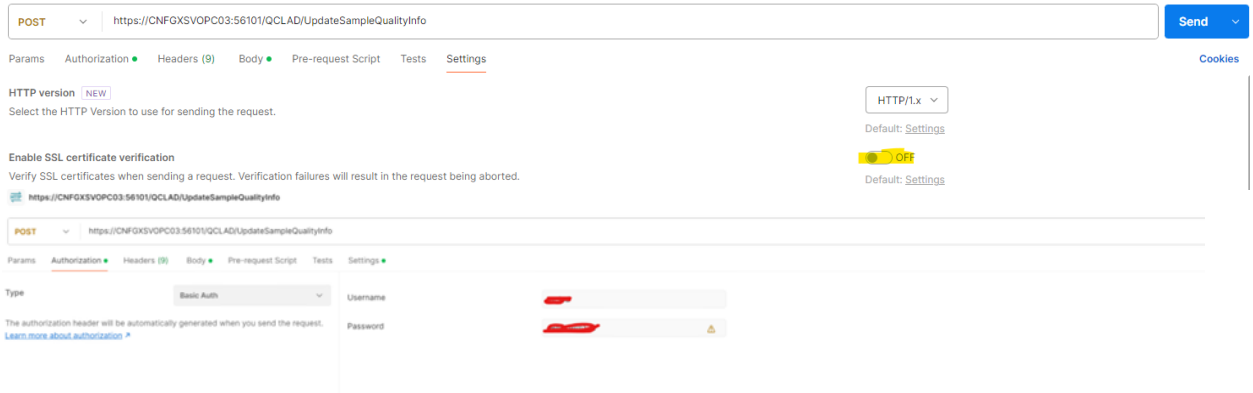
3.2.4.5.1.4 IPQC Sample Info Receiving (QCLAD to MES)

The same as MES interface, please refer to 3.2.2.4.2. IPQC Process Integration.

3.2.4.5.2 API detail

3.2.4.5.2.1 API Authentication Mechanism

API's will use a basic authentication mechanism with username and password.
Below is the sample: (disable SSL Certificate Validation)



3.2.4.5.2.2 API Specifications

API is required to be developed to successfully integrate QCLAD with DCS. API will need to be developed and exposed to DCS, as explained in the table below.

#	API Name	Description	Hosted On	Type
1	UpdateSampleQualityInfo	Insert the test results	Henkel Plant Network	Push

UpdateSampleQualityInfo

This API is invoked by QCLAD whenever they are starting the the sample for testing

Web service URL to access DCS System:
https://CNFGXSVOPC03:56101/QCLAD/UpdateSampleQualityInfo

Web service Type: POST

JSON structure to pass the results will follow the same used for MES.

Web service output:

API will return an object as return details.

Return Code Info

Return Code	Description	Action By QCLAD
0001	Success	No Action
0003	Duplicate Sample Number	Correct sample number and Resend
0005	Invalid JSON	Correct JSON and Resend

Note: In the event of any network failure, QCLAD should resend the data to ensure successful processing.

Response Examples:


```
{
  "MesSampleId": "37158418993",
  "Status": "003",
  "Message": "Duplicate sample number"
}

{
  "MesSampleId": "37158418993",
  "Status": "006",
  "Message": "Sample inserted but batch is inactive"
}
```


3.2.5 SAP Interface_Adjustment

3.2.5.1 Introduction

This document describes the system functionality for the integration of SFG-FG mapping between QCLAD (SampleManager) and SAP as part of Phase B implementation.

3.2.5.2 Purpose

The purpose of this document is to provide detailed specifications for the sap-QCLAD integration, covering SFG-FG mapping processes, including interface requirements, data mapping, and system logic.

3.2.5.3 Requirements Background

Bulk IDH is same as material IDH

Bulk batch number is same as SAP batch number

They are differentiating the material as bulk and FG based on valuation class(semi finished goods are mentioned as bulk , finished goods are mentioned as FG).

If user receive a material with valuation class as semi finished good, SAP will send it as bulk IDH and its batch number as bulk batch number.

There can be scenarios where QCLAD may/may not have FG IDH(master data will not be available in Substance. In QCLAD only those materials which have inspection plan are available) but surely we will have bulk IDH. So, we need to do this mapping based on dynamic data received from SAP.

3.2.5.3.1 BULK-FG Mapping Process Integration

3.2.5.3.1.1 Interface URL address

QCLAD will expose an API for SAP to send SFG-FG information.

3.2.5.3.1.2 BULK-FG Mapping Info Receiving (SAP to QCLAD)

Example request JSON:


```
[
  {
    "srcSystem": "R93",
    "apiuid": "INP_API_690B7D1CB73E1FE0A089A97405134345",
    "bulkFgPlant": [
      {
        "plant": "0759",
        "fillMatList": [
          {
            "bulkFgUuid": "BLK_FG_320B7D1CB73E1FD09F86E2AE26BA691C",
            "fillingIdh": "2163421",
            "fillingBatch": "FILL1234",
            "fillingProcessOrder": "0970635",
            "bulkMatList": [
              {
                "bulkIdh": "2163422",
                "bulkBatchNumber": "BULK1234",
                "bulkProcessNumber": "0970636",
                "fillBatchAsgDate": "2025-08-22:10-45-59"
              },
              {
                "bulkIdh": "2163423",
                "bulkBatchNumber": "BULK1235",
                "bulkProcessNumber": "0970638",
```

```
            "fillBatchAsgDate": "2025-08-22:10-48-59"
          }
        ]
      },
      {
        "bulkFgUuid": "BLK_FG_320B7D1CB73E1FD09F86E2AE26B7B91C",
        "fillingIdh": "2163431",
        "fillingBatch": "FILL2234",
        "fillingProcessOrder": "0970645",
        "bulkMatList": [
          {
            "bulkIdh": "2163432",
            "bulkBatchNumber": "BULK2234",
            "bulkProcessNumber": "0970646",
            "fillBatchAsgDate": "2025-08-22:11-45-59"
          },
          {
            "bulkIdh": "2163433",
            "bulkBatchNumber": "BULK2235",
            "bulkProcessNumber": "0970648",
            "fillBatchAsgDate": "2025-08-22:11-48-59"
          }
        ]
      }
    ]
  }
  ]
}
```


Example response JSON:


```
[
    {
        "srcSystem": "R93",
        "apiuid": "INP_API_690B7D1CB73E1FE0A089A97405134345",
        "bulkFgPlant": [
            {
                "plant": "0759",
                "fillMatList": [
                    {
```

```
                                "bulkFgUuid": "BLK_FG_320B7D1CB73E1FD09F86E2AE26BA691C",
                                "bulkMatList": [
                                    {
                                        "bulkIdh": "2163422",
                                        "bulkBatchNumber": "BULK1234",
                                        "bulkProcessNumber": "0970636",
                                        "ReturnCode": "0001",
                                         "Description": "Success"
                                    },
                                    {
                                        "bulkIdh": "2163423",
                                        "bulkBatchNumber": "BULK1235",
                                        "bulkProcessNumber": "0970638",
                                        "ReturnCode": "0002",
                                         "Description": "No batch found"
                                    }
                                ]
                            },
                            {
                                "bulkFgUuid": "BLK_FG_320B7D1CB73E1FD09F86E2AE26B7B91C",
                                "fillingIdh": "2163431",
                                "fillingBatch": "FILL2234",
                                "fillingProcessOrder": "0970645",
                                "bulkMatList": [
                                    {
                                        "bulkIdh": "2163432",
                                        "bulkBatchNumber": "BULK2234",
                                        "bulkProcessNumber": "0970646",
                                        "ReturnCode": "0001",
                                         "Description": "Success"
                                    },
                                    {
                                        "bulkIdh": "2163433",
                                        "bulkBatchNumber": "BULK2235",
                                        "bulkProcessNumber": "0970648",
                                        "ReturnCode": "0001",
                                         "Description": "Success"
                                    }
                                ]
                            }
                        ]
                    }
                ]
            }
        ]
```


3.2.5.3.1.3 Error handling

ReturnCode	Description
0001	Success
0002	No Batch Found

3.2.5.3.1.4 Field Mapping

No.	MES Field	Description	QCLAD Field Mapping
1	PlantCode	Plant code	Plantcode
2	BulkMaterialList		
3	BulkIdh	material IDH	sample.Hen_IDH
4	BulkBatchNumber	sap Batch number	sample.BatchName
5	BulkProcessNumber	sap process Number	sample.hen_sap_process_number
6	FillingMaterialList		
7	FillingIdh	FillingIdh	sample.Hen_FillingIdh
8	FillingBatch	FillingBatch	sample.Hen_FillingBatch

3.2.5.3.1.5 System Logic

1. SAP sends bulk material data to QCLAD API endpoint
2. System verifies:
    - Bulk IDH exists in Substance table (mandatory)
    - FG IDH availability (optional)
3. For valid bulk IDH:
    - Creates SFG-FG mapping record
    - Returns success response to SAP
4. For invalid bulk IDH:
    - Logs error in "Pending_Mapping" queue
    - Returns error code to SAP

3.2.5.3.2 Storage Location in Plant Integration

3.2.5.3.2.1 Business Scenario

To support material adjustment functionality in MES integration, QCLAD requires storage location data from SAP to:

- Validate storage locations during QC adjustment
- Provide dropdown options for users when specifying adjustment materials
- Ensure consistency with SAP inventory management

3.2.5.3.2.2 Interface Requirements

Source Data: Storage locations are provided via SAP Plant Master interface (LGORT field)

Field Additions:

QCLAD Table	New Field	Source Field	Description
Plant	StorageLocations	LGORT (Plant Master)	Semicolon-delimited list of valid storage locations

3.2.5.3.2.3 Data Flow

Example request JSON:


```
[
  {
    "srcSystem": "Q93",
    "apiuid": "PLN_API_FE412F97F8EB1FD092F4ECE2B9448FF9",
    "parentPlant": [
      {
        "uuid": "PLN_MAS_FE412F97F8EB1FD092F4ECE2C2C50FF9",
        "werks": "S5P9",
        "lgort": "0001/0003/0005/0011/8001/9001/9002/9003/9004",
        "warehouse": "Y",
        "parentplantcode": "0759",
        "name1": "WH CN SH BF DG",
        "bwkey": "S5P9",
        "kunnr": "00PA00S5P9",
        "lifnr": "00PAY0S5P9",
        "name2": "",
        "stras": "No.10359 Puwei Road,  Jinshan",
        "ort01": "Shanghai",
        "land1": "CN",
        "vLand1": "People's Republic of China",
        "cityc": "",
```

```
            "adrnr": "0002392416",
            "childCrhd": [
              {
                "arbpl": "QC-LAB",
                "verwe": "000",
                "planv": "009",
                "delInd": " "
              },
              {
                "arbpl": "S5P9-123",
                "verwe": "PI0",
                "planv": "009",
                "delInd": " "
              },
              {
                "arbpl": "S5P9-LAB",
                "verwe": "PI0",
                "planv": "009",
                "delInd": " "
              }
            ]
          }
        ]
      }
    ]
```


3.2.5.3.2.4 System Logic

1. SAP sends Plant Master data with LGORT field (Example from Plant Master.txt):
   "lgort": "0001/0003/0005/0011/8001/9001/9002/9003/9004"
2. QCLAD parses and stores locations in Plant table

3.2.5.3.3 Inspection Lot Batch Quantity and Unit Integration

3.2.5.3.3.1 Business Scenario

Enable QCLAD to display and utilize batch quantity/unit from SAP inspection lots for:

- Material adjustment display
- QC decision validation against batch size
- MES interface requirements

3.2.5.3.3.2 Interface Requirements

Source Data: Quantity and Unit information are provided via SAP Inspection Lot interface.

Field Additions:

SAP Field	QCLAD Mapping	Example Value
losmenge	Sample.SAP_BatchQuantity	"300" (from inspection lot.txt)
mengeneinh	Sample.SAP_BatchUnit	"KG"

3.2.5.3.3.3 Data Flow

Example request JSON:


```
[
{
  "inspMatPlnt": [
    {
      "werks": "0759",
      "inspmatdata": [
        {
          "uuid": "INSP_DATA_518F395F92E71FD09CD2F5F4CF2E60",
          "matnr": "000000000002101789",
          "prueflos": "14510777",
          "art": "89P",
          "charg": "0006572802",
          "inspLotStatus": "A",
          "aufnr": "",
          "lichn": "",
          "lagortchrg": "",
          "vfdat": "",
          "lwedt": "",
```

```
"qndat": "00000000",
"lifnr": "",
"hsdat": "",
"losmenge": "300",  //batch quatity
"mengeneinh": "KG", //batch unit
"activityInsp": [
  {
    "vornr": "0010",
    "inspecChar": [
      {
        "vorglfnr": "1",
        "sollstpumf": 1,
        "merknr": "10",
        "ersteller": "SINGHA50",
        "erstelldat": "2025-08-06",
        "satzstatus": "1",
        "pmethode": "LOMAPP01",
        "verwmerkm": "00000049"
      },
      {
        "vorglfnr": "1",
        "sollstpumf": 1,
        "merknr": "20",
        "ersteller": "SINGHA50",
        "erstelldat": "2025-08-06",
        "satzstatus": "1",
        "pmethode": "LOMPHV02",
        "verwmerkm": "00000024"
      },
      {
        "vorglfnr": "1",
        "sollstpumf": 1,
        "merknr": "30",
        "ersteller": "SINGHA50",
        "erstelldat": "2025-08-06",
        "satzstatus": "1",
```

```
                    "pmethode": "LOMTAK01",
                    "verwmerkm": "L59"
                }
            ]
        }
    ]
  }
 ]
}
],
"srcSystem": "R93",
"apiuid": "INSP_LOT_API_518F395F92E71FD09CD2F6F823D"
}
]
```


3.2.5.3.3.4 System Logic

1. Receive inspection lot data from SAP
2. Store batch quantity/unit in sample record
3. Utilize in adjustment